INTERIM MILITARY SPECIFICATION

SHIP CONTROL CONSOLES AND COMBINATION EQUIPMENTS

(NAVAL SHIPBOARD USE)

1. SCOPE

1.1 Scope. - This specification covers ship control consoles, steering control consoles, and ship control combination instruments for transmitting and receiving orders and indications for ship control purposes in Naval ships.

1.2 Classification. - The types of ship control consoles, steering control consoles, and ship control combination instruments shall be designated in the following form, as applicable (see 6.1):

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1.2.1 Basic indicator. - The basic indicator for ship control consoles, steering control consoles and ship control combination instruments.

1.2.2 Type of instruments - The type of instruments shall be designated by the following number symbols:

Symbols	Type of Instruments	Details
11 - 29	Ship control combination instruments	Table I
30 - 49	Ship control consoles	Table II
50 - 69	Steering control consoles	Table III

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Symbol	Instrumentation
11	1MB - Engine order, transmitter and indicator 2MB - Engine order, transmitter and indicator LA - Course-steering, double transmitter N - Rudder angle, indicator K - Propeller revolution, indicator
12	1MB - Engine order, double indicator (order and ans.) 2MB - Engine order, double indicator (order and ans.) N - Rudder angle, indicator 1K - Propeller revolution, indicator } One instrument 2K - Propeller revolution, indicator 3K - Propeller revolution, indicator } One instrument 4K - Propeller revolution, indicator Y - Underwater log, indicator
13	1MB - Engine order, double indicator (order and ans.) 2MB - Engine order, double indicator (order and ans.) 1N - Rudder angle, indicator 2N - Rudder angle, indicator } One instrument 3N - Rudder angle, indicator 1K - Propeller revolution, indicator } One instrument 2K - Propeller revolution, indicator 3K - Propeller revolution, indicator } One instrument 4K - Propeller revolution, indicator Y - Underwater log, indicator
14	1MB - Engine order, transmitter and indicator 2MB - Engine order, transmitter and indicator M - Propeller order, triple transmitter-indicator

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Table II - Ship control consoles.

Symbol	Instrumentation
30	1MB - Engine order, transmitter and indicator 2MB - Engine order, transmitter and indicator M - Propeller order, triple transmitter-indicator Speedlight control circuit
31	MB - Engine order, transmitter and indicator M - Propeller order, triple transmitter-indicator Speedlight control circuit
32	1MB - Engine order, transmitter and indicator 2MB - Engine order, transmitter and indicator 4MB - Propulsion control order, switch and indicator lights 1K - Propeller revolution, indicator 2K - Propeller revolution, indicator Clutch position indicators Throttle controls - port and starboard Propeller pitch controls - port and starboard Engine control air pressure gage

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Table III - Steering control consoles.

Symbol	Instrumentation
50	L - Rudder order, transmitter N - Rudder angle, indicator } One instrument Helm angle, indicator (mechanical) LM - Magnesyn compass, indicator LB - Steering emergency switch LC - Ship's course indicator Cable selector switch LCS - Electrical steering - potentiometer type
51	L - Rudder order, transmitter N - Rudder angle, indicator } One instrument Helm angle indicator (mechanical) LM - Magnesyn compass, indicator LB - Steering emergency switch LC - Ship's course indicator LA - Course-steering, double indicator Cable selector switch LCS - Electrical steering - potentiometer type
52	L - Rudder order, transmitter N - Rudder angle, indicator } One instrument Helm angle indicator (mechanical) LM - Magnesyn compass, indicator LB - Steering, emergency switch LC - Forward ship's course indicator LC(A) - After ship's course indicator Cable selector switch LCS - Electric steering - potentiometer type

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Table III - Steering control consoles. (Cont'd.)

Symbol	Instrumentation
53	L - Rudder order, transmitter 1N - Rudder angle, indicator 2N - Rudder angle, indicator 3N - Rudder angle, indicator } One instrument Helm angle indicator (mechanical) LM - Magnesyn compass, indicator LB - Steering, emergency switch LC - Forward ship's course indicator LC(A) - After ship's course indicator Rudder angle transfer switch LCS - Electrical steering
54	L - Rudder order, transmitter N - Rudder angle, indicator } One instrument Helm angle indicator (mechanical) LM - Magnesyn compass, indicator LB - Steering, emergency switch LC - Ship's course indicator 2GR - Course-to-steer indicator Cable selector switch LCS - Electrical steering-potentiometer type

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1.2.3 Mounting and construction. - The type of mounting and construction of the enclosure shall be designated by a single letter symbol as shown in table IV.

Table IV - Enclosure, mounting, and construction

Symbol	Mounting	Construction
B	Bulkhead	Cast
P	Panel	Cast
S	Pedestal	Cast
U	Universal	Cast
D	Deck	Sheet Metal
W	Bulkhead	Sheet Metal
X	Panel	Sheet Metal
Y	Pedestal	Sheet Metal
Z	Universal	Sheet Metal

Note. - Cast enclosures are normally aluminum. Sheet metal enclosures are normally made of sheet brass, except where material scarcity compels the use of sheet steel.

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1.2.4 Additional features.- Additional features and special details shall be designated by a number ranging from 01 through 99 reserved for this purpose in each of tables V, VI, and VII. Tables V, VI, and VII are associated with tables I, II, and III, respectively.

Table V - Additional features and special details of ship control combination instruments.

Symbol	Additional features and special details of ship control combination instruments
01	Three-point mounting (bulkhead) Cast enclosure (aluminum) Condensation drain plug Appropriate identification plates One master dimming rheostat for instrument illumination Operating knob for 1MB transmitter Operating knob for 2MB transmitter Operating knobs (two) for course-steering transmitters Backing indicator light for each shaft (four shafts) Type 5G synchros shall be used for each transmitter Type 23TR6A synchros shall be used for each indicator
02	Three-point mounting (bulkhead) Cast enclosure (aluminum) Condensation drain plug Appropriate identification plates One master dimming rheostat for instrument illumination Backing indicator light for each shaft (four shafts) Type 23TR6A synchros shall be used for each directly positioned pointer Type 3F synchros shall be used for each pointer positioned through gearing.

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Table V - Additional features and special details of ship control combination instruments. (Cont'd.)

Symbol	Additional features and special details of ship control combination instruments
03	Pedestal mounting (eight point) Cast enclosure (aluminum) Condensation drain plug Appropriate identification plates Access panel - (Back of center section) Three dimming rheostats for instrument illumination Operating handle for 1MB transmitter Operating handle for 2MB transmitter Attention-signal push switch for each engine order transmitter Pilot light for power to each engine order transmitter Special 8 '' dial for use with each engine order circuit Type IC/B1S4 bell to announce engine order answers - one for each of circuits 1MB and 2MB Operating knobs for propeller order transmitters (3 knobs) Attention-signal push switch for propeller order transmitter Type IC/Z1S4 buzzer to announce propeller order answers Type 5G synchros shall be used for each transmitter Type 23TR6A synchros shall be used for each indicator

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Table VI - Additional features and special details of ship control consoles.

Symbol	Additional features and special details of ship control consoles
01	Enclosed deck-type mounting Sheet metal enclosure (brass) Cast enclosure (aluminum) Frontal access plate Appropriate identification plates Four dimming rheostats for instrument illumination Operating handle for 1MB transmitter Operating handle for 2MB transmitter Attention-signal push switch for each engine order transmitter Pilot light for power to each engine order transmitter Special 8" dial for use with each engine order circuit Type IC/B1S4 bell to announce engine order answers - one for each of circuits 1MB and 2MB Operating knobs for propeller order transmitters (three knobs) Attention-signal push switch for propeller order transmitter Type IC/Z1S4 buzzer to announce propeller order answers - (one buzzer) Operating knob for speedlight circuit control switch Operating knob for speedlight signal selector switch Handpulse push switch for speedlight circuit Indicator lights, marked "R" and "W", for speed-light circuit Type 5G synchros shall be used for each transmitter Type 23TR6A synchros shall be used for each indicator
02	Same features as Symbol 01 except type 6G synchros shall be used for the engine order transmitters, in lieu of 5G synchros.

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Table VI - Additional features and special details of ship control consoles. (Cont'd.)

Symbol	Additional features and special details of ship control consoles
03	Same features as Symbol 01 except there will be but one engine order circuit (Ckt. 1MB), in lieu of circuits 1MB and 2MB. Only one operating handle shall be provided for the engine order transmitter and shall be located at the center of the drum-shaped enclosure for the engine order instruments. Orders and answers for engine order circuit 1MB will appear on both dial faces. One 5G synchro shall be used for the engine order transmitter. Two 23TR6 synchros shall be used for the engine order answer indication. Three dimming rheostats shall be used in lieu of four as indicated under Symbol 01.
04	Enclosed deck-type mounting Sheet metal enclosure (brass) Frontal access plate Grab rail (removable) Frontal toe space Two cover lifting handles (upper section) One master dimming rheostat for upper section One master dimming rheostat for lower section Operating knob for 1MB transmitter Pilot light for power to 1MB circuit Operating knob for 2MB transmitter Pilot light for power to 2MB circuit Attention-signal push switch for each engine order transmitter. Clutch position indicator lights, marked "In" and "Out", for each shaft (two shafts) Propulsion control order indicator lights, marked "ER", and "PH", for each shaft (two shafts) Pilot light for power to each propeller pitch control, port and starboard.

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Table VI - Additional features and special details of ship control consoles. (Cont'd.)

Symbol	Additional features and special details of ship control consoles
04 (cont'd.)	Type IC/1S4 bell for each engine order answer (Ckt. 1MB and 2MB) - shall be mounted internally Type 5G synchros shall be used for the engine order transmitters Type 23TR6A synchros shall be used for the propeller revolution and the engine order indicators.

Table VII - Additional features and special details of steering control consoles.

Symbol	Additional features and special details of steering control consoles
01	Enclosed deck-type mounting Sheet metal enclosure (brass) Frontal toe space Cover lifting handles - two 12 inch spoke-type steering wheel Frontal access plates One master dimming rheostat for instrument illumination Lever-type emergency steering switch Operating knob for rudder order transmitter Pilot light for rudder order circuit power Push switch for rudder order, circuit "L". Pilot light for steering system power supply Type 5G synchro shall be used for rudder order transmitter Type 23TR6A synchros shall be used for all indicators except as indicated

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Table VII - Additional features and special details of steering control consoles. (Cont'd.)

Symbol	Additional features and special details of steering control consoles
02	Same features as Symbol 01 except add: Switch for circuit "LA" power supply IC/B1S4 bell for circuit "LA" attention-signal push switch for circuit "LA" acknowledgement signal indicator light for circuit "LA" in parallel with audible signal. A type 5F synchro shall be used for positioning the ring dial of the course steering indicator.

1.2.5 Manufacturer. - The number indicating the additional features and special details shall be followed by a dash and a number indicating the manufacturer. This number will be assigned by the bureau or agency concerned at the time of action on drawings of the component.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, standards, and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

NN-B-591 - Boxes, Fiberboard, Wood-Cleated, (for Domestic Shipment).
NN-B-601 - Boxes, Wood-Cleated-Plywood, for Domestic Shipment.
NN-B-621 - Boxes, Wood, Nailed and Lock Corner.
QQ-S-692 - Steel, Sheet, Cold-Rolled.
QQ-S-781 - Strapping, Flat, Steel.
WW-T-799 - Tubing, Copper, Seamless (for Use With Solder-Joint or Flared-Tube Fittings).
LLL-B-631 - Boxes, Fiber, Corrugated (for Domestic Shipment).
LLL-B-636 - Boxes, Fiber, Solid (for Domestic Shipment).

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MILITARY

JAN-P-104 - Packaging and Packing for Overseas Shipment - Crates, Sheathed, Wood, Nailed.
JAN-P-105 - Packaging and Packing for Overseas Shipment - Boxes, Wood, Cleated, Plywood.
JAN-P-106 - Packaging and Packing for Overseas Shipment - Boxes, Wood, Nailed.
JAN-P-108 - Packaging and Packing for Overseas Shipment - Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.
MIL-P-116 - Preservation, Methods of.
JAN-P-132 - Packaging and Packing for Overseas Shipment - Crates; Unsheathed, Wood; Nailed (for Maximum Net Load of 2,500 Pounds).
JAN-P-139 - Packaging and Packing for Overseas Shipment - Plywood, Container Grade.
MIL-B-233 - Boxes, Repair Parts.
JAN-P-658 - Packaging and Packing of Electrical Equipment and Spare Parts (Electronic, Electrical, and Electro-Mechanical).
MIL-I-983 - Interior Communication Equipment, General Specification for.
MIL-B-994 - Brass, Naval, Wrought.
MIL-F-1183 - Fittings, Tube, Bronze, Cast (Silver Brazing).
MIL-S-2335 - Synchros, 60 Cycle, 115 Volt.
MIL-C-6057 - Crates and Boxes; Aircraft and Airframe Component Parts (General Specification for).
MIL-L-10547 - Liners, Case, Waterproof.
MIL-I-15024 - Identification Plates, Information Plates and Marking for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-B-15071 - Books, Instruction; Preparation, Contents, and Approval.
MIL-A-15303 - Audible Signals, Interior Communication and Small Craft (Bells, Buzzers, Horns and Sirens) (Naval Shipboard Use).
MIL-C-16238 - Control Equipment, Propeller Pitch.
MIL-T-16784 - Terminal Boards.
MIL-A-17129 - Aluminum Alloy Casting; Sand, Resistant to Salt-Water.
MIL-S-17299 - Ship Control Order and Indicating Equipment - Self-Synchronous.

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NAVY DEPARTMENT
General Specifications for Inspection of Material
46B24 - Bronze: Castings, Ornamental

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking of shipments

DRAWINGS

BUREAU OF SHIPS

S2407-533062	Magnesyn Compass System, Circuit LM; Cable Assembly for Transmitter Indicator
9000-S6202-73193	Contact Makers - Panel Mounting - Multiple Rotary
9000-S6504-73687	Standard Dial Markings for I. C. Telegraphs and Indicating Systems
9000-S6202-73754	Switch Box - W. T. (Single), Type E-16B
9000-S6405-73786	Speedlight Controller, Circuit and Dial Arrangment for Use in Ship's Control Console Unit
9000-S6202-73826	Switch, Multipole Rotary, Panel Mounted - 10 Amp., 120 Volt A. C. - Type S-2JR
9000-S6504-74055	SwitchS-Lever Operated - Base Mounting-SBM Enclosed
9000-S6202-73907	Switchboard and Indicator Light, 2 Lamp, SPF
9-S-4222-L	Lamps, Electric, Used on Naval Vessels
9-S-5235-L	Stuffing Tubes for Electric Cables Entering-Sheet Metal Enclosures
S4105-3, 119, 768	Single Pointer Air Gage
S6505-1, 382, 705	Steering Control Console
S6505-1, 382, 706	Ship Control Console
S6505-1, 383, 195	Ship Control Console with Variable Propeller Pitch Controls
S6505-1, 412, 982	Ship Control Indicator
S6505-1, 430, 783	Ship Control Indicator Transmitter

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the cont racting officer.)

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2.2 Other publications - The following publication, of the issue in effect on date of invitation for bids, unless otherwise stated, forms a part of this specification:

CONSOLIDATED FREIGHT CLASSIFICATION COMMITTEE
Consolidated Freight Classification - Ratings, Rules and Regulations

(Application for copies should be addressed to the Consolidated Freight Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Qualification. - The ship control consoles and combination equipments furnished under this specification shall be a product which has been tested and has passed the qualification tests specified in section 4 (see 6.2).

3.2 General features. -

3.2.1 The equipment shall be in accordance with the following paragraphs of Specification MIL-I-983 in addition to the requirement specified herein. Whenever a requirement of Specification MIL-I-983 conflicts with a requirement of this specification, the requirement of this specification shall govern.

Materials, general
Stressed or vital parts
Unstressed or non-vital parts
Wood
Magnesium
Aluminum
Brass and bronze
Iron and steel
Protection against corrosion (salt spray test is required)
Gaskets
Component dial windows
Plastics
Inflammable materials
Toxic compounds
Lubricants and lubrication
Electrical parts - general
Electron tubes
Electron tube and/or capacitor sockets
Capacitors
Resistors
Potentiometers
Transformers
Relays
Synchros (see 3.4.6)
Fungus-inert materials
Restriction of materials
Electrical tapes
Enclosures - general
Accessibility
Degree of enclosure
Enclosure mounting
Stiffening grooves
Minimum sheet thickness
Through bolting
Terminal tubes
Condensation drain
Cover bolts
Ventilation

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Size (for submarines)
Temperature and humidity (special cold limit applies)
Threaded parts
Locking devices
Washers
Interchangeability and standardization
Shock, vibration and inclination (vital vibration test required)
Component dials and pointers
Painting
Designation and marking
Design for production
Rounded corners and edges
Internal unit protection
Ball bearings
Drilled and tapped holes
Welding
Primary power circuits
Power supply tolerances (normal power supply - 115 volts 60 cycles a.c. single phase)
Personnel protection
Clearances (insulation resistance and dielectric strength)
Shielding and radio frequency noise interference
Audible noise (Airborne only)
Grounds
Soldering
Mounting of parts
Internal assembly connection (see 3.4.2)
Terminal strips, connectors and terminals
Wiring
Dial illumination (see 3.4.8)
Lamps
Batteries
Drawings
Instruction books
Maintenance parts kit
Repair parts
Workmanship and general inspection
General information

3.2.2 Special material features. -

3.2.2.1 Push switches. - Push switches may incorporate internal units in accordance with Drawing 9000-S6202-73754; or shall be of a design satisfactory to the bureau or agency concerned, employing sensitive switches.

3.2.2.2 Electrical tapes. - The use of electrical tapes shall be held to a minimum.

3.2.2.3 Heads of screws and bolts. - Heads of screws and bolts which hold parts or subassemblies on the back of an assembly or compoent panel shall have a different type of head in order to distinguish them from screw heads and bolt heads used in securing the panel itself.

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3.2.2.4 Identification plates. - Identification plate details shall be in accordance with Specification MIL-I-15024. The Bureau of Ships stock numbers for the ship control indicators, ship control indicator-transmitters, ship control consoles, and steering control consoles shall be obtained as specified in 3.2.2.7. The identification plates shall be located so that they are readily associated with the assembly to be identified.

3.2.2.4.1 Other designation plates. - Designation plates, as required, shall be provided to identify the instruments, switches, controls and other components of the consoles.

3.2.2.5 Wiring diagrams. - An electrical wiring diagram for each equipment shall be provided and mounted inside the respective equipments. The diagrams shall be provided in either nickel-copper alloy plates or plastic plates, and shall be mounted inside the respective equipments in a convenient and accessible location so as to be usable in servicing the equipment.

3.2.2.6 Location of terminal strips. - Terminal strips shall be provided for connection of ships wiring, in a location easily accessible from the access plates. All components mounted on or in the equipment shall be wired to these terminal strips.

3.2.2.7 Bureau of Ships stock numbers. - Bureau of Ships stock numbers for the equipments and associated maintenance parts kits will be furnished to the contractor, by the Bureau of Ships, 45 days after the time of approval of the working plans and upon request of the contractor. These stock numbers shall be applied to the equipments and the maintenance kits prior to shipment by the contractor.

3.2.2.8 Enclosure material. - Material for enclosure cases and covers shall be in accordance with one of the following materials.

3.2.2.8.1 Cast aluminum (class 3 or class 4) shall be in accordance with Specification MIL-A-17129.

3.2.2.8.2 Sheet brass shall be in accordance with Specification MIL-B-994.

3.2.2.8.3 Sheet steel shall be in accordance with Specification QQ-S-692.

3.2.2.8.4 Cast bronze shall be in accordance with Specification 46B24.

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3.2.2.9 Standard parts and materials. - The use of standard parts and materials, covered by specifications and drawings listed in section 2 herein; and also covered by Specifications MIL-I-983 and MIL-S-17299 is mandatory wherever those specifications and drawings are applicable to the parts or materials used.

3.3 Mechanical features. -

3.3.1 Construction. - The design of each equipment shall take into consideration the possible shortages due to emergency conditions, necessitating the use of alternate materials such as steel, and the designs shall permit the use of such an alternate material with a minimum of retooling. Items of equipment installed in the consoles shall consist of as few components, assemblies and subassemblies as practicable.

3.3.1.1 Mounting. - Each of the consoles shall be designed for deck mounting. The mountings for the combination instruments shall be as designated for the individual instruments.

3.3.1.1.1 Shock mounting of the equipment as a unit is not required. However, the individual components shall be inherently shock and vibration proof.

3.3.1.2 Enclosures. -

3.3.1.2.1 Consoles. - Each console shall consist of a brass structural and formed angle framework, enclosed with panels of fabricated or drawn sheet brass, unless otherwise specified in the contract or order. The enclosure of each console shall be brazed or otherwise securely fastened to the framework, except where removable panels are required for access to the instruments and other components. The framework shall be of rugged construction capable of supporting the full weight of the instruments, components and enclosing panels.

3.3.1.2.2 Combination instruments. - The enclosure and cover of each type of combination instrument shall be aluminum casting, unless otherwise specified in the contract or order. The covers of these combination instruments shall be fastened to the cases by means of bolts and shall form a watertight fit against suitable gaskets to preclude entrance of water.

3.3.1.2.3 Enclosure material shall be in accordance with 3.2.2.8.

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3.3.1.3 Cover handles and grab rods. - Cover handles and grab rods for the individual equipments shall be provided as indicated in appropriate instrument sections. In general, all handles shall be 3/4-inch diameter and all grab rods shall be 1-inch diameter. Handles and grab rods shall be in one piece and shall have a clearance of approximately 1-1/2 inches from the surface of the equipment involved.

3.3.1.4 Toe space. - Toe room shall be provided at the base of any console in which there is not sufficient clearance for the toes of personnel when personnel is in the normal operating position. The space shall run the width of the console and shall be approximately 4 inches in height and 3 inches in depth. Stainless steel kick pads shall be provided along the back of the toe space.

3.3.1.5 Accessibility. - Removable panels (in addition to the instrument panel covers) to permit access to equipment and terminal strips (for connection to ship's wiring) installed within the consoles, shall be limited to the front of each console.

3.3.1.6 Fastening removable panels. - All removable panels shall incorporate two pins to hold the panel in place while the fastening screws are being removed. Means shall be provided to keep the fastening screws captive in the removable panel.

3.3.1.7 Instrument mounting. - Unless otherwise specified herein, all instruments shall be mounted in the framework structure from the top. Each instrument shall be self supported.

3.3.1.7.1 Consoles. - The sides and bottom of each instrument framework shall be constructed so as to provide adequate protection to the instrument involved, and shall be covered with sheet brass to provide individual wells for each instrument. Additional frameworks, as necessary, shall be provided to serve as guides when removing or replacing the instruments from the respective consoles, either by providing thumb room or an extracting tool. If an extracting tool is provided it shall be mounted in a clamp within the respective console; but, if possible, the use of an extracting tool shall be avoided.

3.3.1.8 Instrument features, general. - Unless otherwise specified herein, each instrument shall be complete in itself, with dials (fixed or rotating), pointers, dial illumination and other indicating devices attached. The dials, pointers and indicating devices of the respective instruments shall be visible through windows incorporated in the instrument panel covers. The instrument panel covers shall support the glass instrument windows, bezels, control knobs and associated dial

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illumination controls in addition to the switches, push switches and other items or equipment specified herein. Each instrument shall be provided with a mechanical light trap, or baffle, to eliminate the use of rubber or felt for light seals. Supports for the light trap, or baffle, shall not extend beyond the instrument framework in order to avoid damage when the instrument is removed from the supporting framework or well. Each instrument internal assembly shall be so mounted that no internal operating parts are disturbed when the unit is removed from the enclosure. Clamping screws shall be accessible for adjustment of the synchro transmitter(s) and receiver(s) when the cover is raised. Protective extensions shall be provided on the internal assemblies to provide a level mounting when the internal assembly is placed on a work bench in either the upright or inverted position.

3.3.1.9 Location of operating attachments. - All operating knobs, dimmer knobs, signal push switches, and pilot lights, shall be located on the enclosure surface which contains the dial window(s) unless otherwise specified herein.

3.3.1.10 Operating knobs. - The stem of each operating knob shall be provided with a smooth skirt cast integral with the knob and which shall completely cover the gland nut of the shaft stuffing tube, or shall be provided with other means to prevent cutting the operator's knuckles. The operating knob shall clear the surface of the enclosure by at least 0.80 inch, and shall be shaped to provide a firm and positive grip for the operator. The operating knobs shall be of cast bronze in accordance with Specification 46B24.

3.3.1.11 Removable knobs. - Transmitter operating knobs, if removable with the cover, shall have positive means for engaging the drive mechanism when secured in the operating position.

3.3.1.12 Watertight glands. - Watertight glands shall be provided for all operating shafts extending through the enclosures.

3.3.1.13 Pointers. - Pointers shall be of sheet aluminum or sheet brass, or of plastic, depending upon the method used for illumination of the pointers. Where sheet aluminum or sheet brass is used, the pointers shall have rolled or bevelled edges so as to contribute to the reflection of light provided for the pointer illumination by the basic equipment illumination.

3.3.1.13.1 Shape. - Where two or more concentric pointers are incorporated in the equipment, the pointers shall be designed so as to provide ready and positive identification by the differences in their shapes.

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3.3.1.13.2 Color. - The color of the pointers and markings shall be as specified for the individual instruments in following sections.

3.3.1.14 Instrument panel covers. - Each instrument panel cover of the hinged type shall be hinged at the rear of its section of the console to permit the cover to be easily opened for access to the instruments installed in the console; and for access to the illumination lamps, instrument windows, illumination controls, switches, and other items installed on the underside of the cover. Lifting of the instrument panel of the consoles shall be by means of grab rails or cover handles at the left and right sides of the top of the cover. A support arm shall be provided for each instrument panel cover to support the cover in the raised position. Each instrument panel cover shall incorporate a center clamping support to the cover. In addition, each instrument panel cover shall incoporate a spring catch on the cover support which will require deliberate release in order to close the cover. Each instrument panel cover, when closed, shall form a watertight fit against gaskets in accordance with Specification MIL-I-983. The instrument panel covers shall possess adequate rigidity to ensure proper alignment and operation of all items attached to the covers; and bracing, or stiffening members, shall be incorporated in the cover, as necessary, to provide adequate rigidity. Means shall be provided in each console to secure the instrument panel covers in the closed position.

3.3.1.15 Condensation drain. - Condensation drains shall be provided at the lowest point of the bottom side (not the mounting side) of the bulkhead mounted ship control combination instruments. A condensation drain shall be provided in the terminal tube plate with the outlet located on the outer (exposed) side of the terminal tube plate.

3.3.2 Degree of enclosure. - The console units shall be splashproof. The combination instruments shall be watertight.

3.3.3 Compactness of design. - The equipment shall be as compact as practicable, consistent with sound engineering principles of design necessary to meet the requirements of this specification. The limiting dimensions as shown on the reference drawings shall not be exceeded unless approved by the bureau or agency concerned.

3.3.4 Bulkhead mounted equipment with cast enclosures. - These equipments shall be provided with cable entrance holes drilled, tapped, and plugged in accordance with Specification MIL-I-983. Entrance holes shall be provided in the bottom of each cast enclosure as specified herein. Adequate boss areas

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shall be provided in the bottom of each enclosure so as to provide for terminal tubes required for the larger size cables specified. A table of terminal tube sizes is shown on Drawing 9-S-5235-L.

3.3.5 Pedestal mounted equipment with cast enclosures. - Pedestal mounted equipment with cast enclosures shall be provided with a terminal tube plate in accordance with Specification MIL-I-983. Cable entrance holes will be drilled in these terminal tube plates by the installing activity. No cable entrances will be made through the top surfaces of the enclosure. The terminal tube plate shall be provided with adequate area to provide for terminal tubes required for the larger size cable listed for each equipment. A table of required terminal tube sizes is referred to in 3.3.4.

3.3.6 Gearing. - Unless otherwise specified herein, when gearing is used between an operating knob (or handle) and its associated pointer, the motion transmitted shall be in a 1 to 1 ratio.

3.4 Electrical features. -

3.4.1 Terminal boards. - Terminal boards shall be in accordance with Specification MIL-T-16784.

3.4.2 Internal assembly connection. - Unless otherwise specified in the contract or order, each electrical instrument shall have attached thereto permanently mounted plugs to fit into permanently mounted receptacles located in the instrument framework, or well, as specified in 3.3.1.7. Guide pins of sufficient length to insure proper alignment of plugs and receptacles shall be provided. Adapter cables shall be provided in the maintenance kits for connecting an instrument to the respective console when the instrument is removed from the console for servicing.

3.4.3 Adjustment for electrical zero. - Instrument pointers, dials and other indicating devices of synchro operated instruments shall be adjusted for electrical zero. This adjustment shall also be made for instruments included in the maintenance kit. The instruction book shall contain complete information for electrically zeroing instruments.

3.4.4 Fuses and resistors. - All fuses associated with the instruments and controls shall be of the dead front neon indicating type and shall be mounted on a sub-panel within the respective consoles. These sub-panels shall be readily accessible by removal of the front panel of the respective console. All fuses shall be marked to indicate the associated circuit or instrument. All resistors installed in the consoles shall be located so as to be readily accessible for servicing or replacement.

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3.4.5 Wiring. - All instruments, switches, controls and other devices shall be wired in a neat and workmanlike manner to terminal boards inside the respective equipment. The terminal boards shall be located so as to be readily accessible through the access openings of the respective consoles, pedestal mounted equipments, and bulkhead mounted equipment with cast enclosures. Ships wiring will be brought up through the deck under the equipment for connection to the terminals of the terminal boards of the respective equipments. All terminal board terminals shall be marked to identify the circuit involved. Grouping of terminals shall be logical and repair terminals provided where necessary to facilitate future arrangements when required. In each console, a minimum of ten repair terminals shall be provided.

3.4.5.1 Wiring to instruments. - The wiring between terminal boards and the permanently mounted receptacles specified in 3.4.2 shall be arranged so that removal of an individual instrument assembly and its associated plug and receptacle assembly will not affect the operation, servicing or removal of any other instrument assembly incorporated in the equipment.

3.4.6 Synchros. - Synchros shall be in accordance with Specification MIL-S-2335.

3.4.6.1 Loading characteristics of synchro units. - The maximum loading of a type 5G synchro transmitter shall be limited to (a) twelve type 1F or 23TR6 synchro receivers; or (b) nine type 3F or 31TR6 synchro receivers; or (c) six type 5F synchro receivers; or (d) appropriate combinations of type 1F or 23TR6, 3F or 31TR6, and 5F synchro receivers. The maximum loading of a type 6G synchro transmitter shall be limited to (a) twenty-four type 1F or twenty type 23TR6 synchro receivers; or (b) sixteen type 3F or twelve type 31TR6 synchro receivers; or (c) twelve type 5F synchro receivers; or (d) appropriate combinations of type 1F or 23TR6, 3F or 31TR6, and 5F synchro receivers. The maximum loading of a type 37TX4 synchro transmitter shall be limited to (a) twelve type 23TR4 synchro receivers; or (b) nine type 31TR4 synchro receivers; or (c) six type 37TR4 synchro receivers; or (d) appropriate combinations of type 23TR4, 31TR4, and 37TR4 synchro receivers.

3.4.7 Audible signals. - Audible signals shall be in accordance with Specification MIL-A-15303. Type IC/B1S4 bells and type IC/Z1S4 buzzers shall be used.

3.4.7.1 Location of audible signals. - Audible signals shall be mounted on the inside of consoles and on the outside of pedestal mounted equipment unless otherwise specified in the contract or order. All console mounted audible signals shall be located so that they can be serviced from the front access plate.

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3.4.8 Instrument dial illumination. - Dial illumination, in accordance with Specification MIL-I-983, shall be provided for the dials of individual equipments as follows:

3.4.8.1 The instrument dials shall be illuminated by the "Duo-panel" system, a special form of edge illumination in which a translucent plastic outer panel (dial) is substantially isolated from intimate surface contact (optical contact) with a transparent plastic light-conducting panel.

3.4.8.2 The outer panel shall present a conventional daylight appearance, i.e., its front surface shall have "white" markings, numerals, etc. on an opaque dull black background. While the (diffuse) daylight reflectivity of the "white" markings shall be sufficiently high to result in adequate contrast with the background and good resulting legibility, the (diffuse) transmission of these markings shall be sufficiently high to permit the successful use of the outer panel in the "Duo-panel" system.

3.4.8.3 The outer panel shall not exceed 1/16 inch in thickness, and shall be: (a) of a thermo-setting plastic material such as a translucent, graphic, melamine-formaldehyde impregnated, paper-base laminate, or (b) of a high softening temperature acrylic material, having engraved (depressed) or embossed (relief) markings, and having an opaque black background, applied over a translucent white undercoating. The rear surface of this panel may be embossed with a screen or pebbled pattern to permit physical supporting contact while reducing to unimportance the areas of optical contact between it and the associated light conducting panel, if the use of spacers is not desirable.

3.4.8.4 The light conducting panel shall be of 1/4 inch thickness, high softening temperature, colorless, transparent, acrylic sheet, to the rear surface of which small areas of compatible white lacquer, or enamel, have been applied, as necessary, to permit light to escape from the panel and illuminate the markings on the outer panel. This paint shall be selectively applied by any convenient silk screening or stenciling means capable of controlling the areas of the individual paint dabs in accordance with a pre-determined, experimentally derived pattern known to result in uniform illumination of all dial markings.

3.4.8.5 Light shall be obtained from lamps (see 3.4.8.8) so located as to introduce light into the conducting panel, and provided with individual cylindrical red plastic (high temperature) filters of spectral transmission characteristics as specified in Specification MIL-I-983. These filters shall be so designed that there is no escape of white light into any part of the illuminated dial or instrument.

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3.4.8.6 Instrument pointers shall be either (a) of metal in accordance with Specification MIL-I-983, or (b) of a thermo-setting plastic material or acrylic material referred to in 3.4.8.3.

3.4.8.6.1 Metal pointers, when used, shall be illuminated by light distributed across the face of the dial at angles approaching the parallel to the plane of the dial. This light may be distributed by reflection from the inner surface of a cylindrical or rectangular metal accessory, from surfaces of the enclosure or by an auxiliary plastic illuminator assembled above, but close to, the surface of the outer dial. Baffles, or louvers, and bezels shall be provided to reduce to a minimum all glare occasioned by the auxiliary pointer illumination, and to entirely prevent direct view of any lamp filament, reflected image of a lamp filament, luminous surface of a plastic illuminator, or other disturbingly bright object.

3.4.8.6.2 Plastic or acrylic pointers, when used, shall be illuminated by light conducted through the pointer hubs from the light conducting inner panel, thus affording an indirect pointer illumination.

3.4.8.7 The means of illuminating the pointer should utilize light from the same lamps that are used to illuminate the dial, but, if necessary, and particularly in larger instruments, separate lamps may be used as required. The lamps shall be in accordance with Drawing 9-S-4222-L.

3.4.8.8 The lamps shall be energized from the secondary of a transformer, and the lighting circuit shall be equipped with a rheostat, or other control device, to control light intensity from maximum value to full extinction when all lamps, or 50 percent of the lamps are operative. At 115 volts input to the transformer, and with all lamps operative, the lamp socket voltages shall not exceed the following values calculated for 1000-hour life:

Navy type	Volts
TB11	7.5
TB14	6.6
TS53	6.87
TS52	6.87

3.4.8.8.1 The transformer shall have minimum voltage regulation consistent with reasonable compactness.

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3.4.8.9 Instrument illumination power supplies. - Dial illumination shall be energized from power supplies of certain instruments and controls as hereinafter specified. Variable resistors shall be connected on the low voltage side of illumination transformers.

3.4.8.10 Illumination controls. - One instrument illumination control shall be provided for each section of each control. These controls shall be provided with an "OFF" position.

3.4.9 Accuracy. - All indicator pointers, indicator rotating dials, etc. shall follow the associated transmitter setting within one geometrical degree for all variation of supply line voltage and frequency when tested in accordance with 4.3.2.

3.4.10 Oscillations. - When tested as specified in 4.3.3, the indicator pointers or indicator dials shall come to rest at a position synchronous with the transmitter within three seconds.

3.4.11 Pilot lights. - Each pilot light shall be visible through a small watertight window in the instrument enclosure. The color of the light, as seen by the operator, shall be red in accordance with Specification MIL-I-983. The pilot light shall have sufficient intensity of brightness to be seen under daylight operating conditions, but shall not be objectionably bright when seen under night operating conditions (darkened spaces). A transformer shall be used in conjunction with one of the following Navy type lamps:

TB11
TB14
TS130
TS131
TS132

The lamps shall be energized from the secondary winding of the transformer. The primary winding of the transformer shall be connected across the power supply of the synchro transmitter designated for the applicable instrument.

3.5 Ship control consoles. -

3.5.1 Type IC/31D03. - This console shall be arranged in general accordance with Drawing S6505-1, 382, 706. The console shall contain the instruments and controls for transmitting engine orders, for transmitting propeller revolution orders, and for controlling the ship's aircraft warning and speed lights.

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3.5.1.1 Console instrumentation. - The console shall consist of two sections, an upper or drum section and a lower section.

3.5.1.1.1 Upper or drum section. - The upper or drum section shall contain the following equipment:

(a) Engine order indicator-transmitter.
(b) Pilot light for (a).
(c) Push switch for (a).
(d) One operating handle for (a).
(e) Dimming rheostat for illumination of (a).
(f) Speed light circuit control switch.
(g) Speed light signal selector switch.
(h) Operating knob for each of (f) and (g).
(i) Indicator lights for speed light system.
(j) Hand pulse push switch for speed light.

3.5.1.1.2 Lower section. - The lower section shall contain the following equipment:

(a) Propeller order indicator-transmitter.
(b) Push switch for (a).
(c) Operating knobs for (a).
(d) Type IC/Z1S4 buzzer for (a).
(e) Type IC/B1S4 bell for (a) of 3.5.1.1.1.
(f) Dimming rheostat for illumination of (a) and also for speed light controls, (f) and (g) of 3.5.1.1.1.

3.5.1.2 Engine order indicator-transmitter. - This indicator-transmitter shall provide a means for transmitting engine orders and receiving acknowledgements of engine orders between the console and the engine room control station. This indicator-transmitter shall contain the mechanism necessary to transmit engine orders (and indicate the orders transmitted), and to indicate the replies (acknowledgements) received. The transmitter shall be positioned by means of one operating handle located in the middle of the drum section and with an axis of rotation which coincides with the axis of the dials and the drum. The handle is not as shown on Drawing S6505-1, 382, 706. This handle shall contain a pointer on each end to indicate on the dial faces the order transmitted. The acknowledgement of an engine order shall be indicated on both the dial faces of the drum section (one type 23TR6A synchro shall be used for each dial).

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3.5.1.2.1 Dial markings. - Dial markings, lettering, and graduations shall be in accordance with Drawing 9000-S6504-73687 (figure 1 for right hand dial and figure 2 for left hand dial). Dials shall be 8 inches in diameter in accordance with Specification MIL-I-983. The dial marking shall be white against a black background.

3.5.1.2.2 Pointers. - The pointers showing orders shall be incorporated with the operating handle. The pointers indicating acknowledgement of engine orders shall be located within the dial windows and shall be white.

3.5.1.2.3 Synchros. - A type 23TR6A synchro unit shall be used to position the acknowledgement pointer of each dial. A type 5G synchro shall be used for transmitting the engine order. Each synchro unit shall be adjustable in position, and shall be so adjusted in position that the associated pointer indicates the center of the "STOP" position on the dial when the synchro unit is at electrical zero.

3.5.1.2.4 Operating handle. - In addition to the features as indicated in 3.5.1.2, the operating handle shall contain a means of retaining position at a given order and shall require a positive action to disengage the handle when it is desired to move it to a new position.

3.5.1.2.5 Detents. - The transmitter generator operating shaft shall be fitted with a detent wheel to insure stability of position and to operate electric contacts in an audible signal circuit. (The audible signal circuit is used to attract attention of engine room personnel to a change in engine orders.) The detent wheel and the signal contacts shall insure proper operation (signal contacts shall close and remain closed for a sufficient period of time) of the audible signals when the transmitter operating handle (or knob) is moved rapidly from one position to another. (In knob operated transmitters, the detent wheel shall be provided with a stop at a point corresponding to the division line between the bottom segment on the dial and the adjacent blank segment.)

3.5.1.2.6 Signal push switch. - In addition to the signal contacts specified in 3.5.1.2.5, a watertight manually operated push switch, in accordance with 3.2.2.1, shall be provided for the audible signal circuit emanating from the engine order indicator-transmitter. The push switch shall be electrically connected in the circuit in parallel with the signal contacts specified in 3.5.1.2.5, in order to permit "at will" signalling over the audible signal circuit. The signal push switch shall be connected across terminals MB/T and MB7.

3.5.1.2.7 Pilot light. - The indicator-transmitter shall be provided with a pilot light, in accordance with 3.4.11, visible through an aperture in the instrument case. The power supply for this pilot light shall be taken from the power leads for the transmitting synchro unit associated with this instrument.

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3.5.1.2.7.1 This light shall be in accordance with type B-27A of Drawing 9000-S6202-73907, modified so that the top of the cap projects not more than 1/4 inch above the panel. The marker disc shall have a 1/8 inch diameter hole in the center.

3.5.1.2.8 Illumination power supply. - Dial illumination of this instrument shall be energized from the power supply of the indicator portion of the instrument.

3.5.1.2.9 Illumination control. - One instrument illumination control rheostat shall be provided for this equipment. The control shall be provided with an "OFF" position.

3.5.1.2.10 Terminal markings. - Terminals marked MB/T, MBB/T, MB1, MB2, MB3, MB/I, MBB/I, MB4, MB5, MB6, MB7, and MB8 shall be provided on the terminal strips for the engine order circuit.

3.5.1.2.10.1 Terminals for audible signal. - Terminals MBB/I and MB8 shall be used to provide connections for an audible signal. The audible signal shall be provided by a type IC/B1S4 bell located on the lower section of the console (see item (e) of 3.5.1.1.2). The type IC/B1S4 bell will be energized by means of a signal switch in the engine room indicator-transmitter (two type IC/B1S4 bells are shown on Drawing S6505-1,382,706).

3.5.1.3 Speed light controller. - This system shall provide means for controlling the aircraft warning and speed lights. The system shall be wired and arranged as shown on Drawing 9000-S6405-73786. Paragraphs 3.2.2.1, 3.3.1.9, 3.3.1.10, 3.3.6, and 3.4.9 shall apply.

3.5.1.3.1 Indicator lights. - The indicator lights shall be red lights of a red color in accordance with Specification MIL-I-983. The right hand light shall incorporate a black "W" in the bull's eye. The left hand light shall incorporate a black "R" in the bull's eye. Indicator lights shall be in accordance with the units as indicated on Drawing 9000-S6405-73786.

3.5.1.3.2 Dials. - Dial marking shall be as shown on Drawing 9000-S6405-73786. The dial markings shall be white on a black background. The dials shall be 4 inches in diameter in accordance with Specification MIL-I-983.

3.5.1.3.3 Pointers. - A single, white pointer with a matte finish shall be provided for each switch dial

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3.5.1.3.4 Dial windows. - Dial windows and their associated gaskets shall be in accordance with Specification MIL-I-983.

3.5.1.3.5 Switches - The rotary switches for the circuit control switch and the signal selector switch shall be as indicated on Drawing 9000-S6405-73786.

3.5.1.3.6 Terminal strip marking - The terminal strip shall be marked as shown on Drawing 9000-S6405-73786.

3.5.1.3.7 Illumination - The illumination circuit shall obtain its power supply from the indicator side of the engine order system. Illumination shall be in accordance with 3.4.9.

3.5.1.4 Propeller order indicator-transmitter - This indicator transmitter shall provide a means for transmitting and indicating propeller orders from the ship control console, and for receiving and indicating the acknowledgement of the propeller order. Paragraphs 3.2.2.1, 3.2.2.8, 3.3.1.10 and 3.4.9 shall apply.

3.5.1.4.1 Dials - This instrument shall be fitted with three revolving type dials for the transmitter, a separate dial for "units", "tens", and "hundreds", and three similar dials for the indicator.

3.5.1.4.1.1 Dial numerals - Numerals "0" to "9" inclusive, shall be arranged around the dial in a counterclockwise direction for increasing values. The dials for the transmitter and indicator shall be arranged in accordance with figures 12 and 11, respectively, of Drawing 9000-S6504-73687. These numerals shall be not less than 1/2 inch in height. The style of the lettering shown for the numerals on Drawing 9000-S6504-73687 shall apply.

3.5.1.4.2 Instrument windows - This instrument shall be provided with three windows for the transmitter and three windows for the indicator; in each instance, one each for "units", "tens" and "hundreds". The size of the windows shall be such as to display only a single numeral, and exclude display of either adjacent numeral on the revolving type dial. The arrangement of the instrument windows shall be in general accordance with the window arrangement shown in conjunction with figures 11 and 12 of Drawing 9000-S6504-73687. The word "indicator" shall be shown immediately above the window openings associated with the indicator dials; and the word "transmitter" shall be shown immediately below the window openings associated with the transmitter dials.

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3.5.1.4.3 Synchros. - Type 23TR6A synchro units shall be used for positioning the three revolving type dials of the indicator. Type 5G synchro units shall be used for transmitting the desired propeller order, and shall be positioned in association with the three revolving type dials of the transmitter. Each synchro unit shall be adjustable in position, and shall be so adjusted in position that the associated revolving type dial indicates the numeral "0" when the synchro unit is at electrical zero.

3.5.1.4.4 Operating knobs. - Each type 5G synchro unit and its associated revolving type dial shall be positioned by means of an operating knob. Clockwise rotation of the operating knob shall result in clockwise rotation (for increasing values) of the associated revolving type dial.

3.5.1.4.4.1 Location. - Each operating knob shall be located below the instrument window, and directly below the center of the respective instrument window.

3.5.1.4.5 Detents. - The operating shaft for each transmitter (type 5G synchro unit) shall be fitted with a detent wheel to insure stability of position, and to operate electric contacts in an audible signal circuit. (This audible signal circuit terminates in a type IC/B3S4 bell associated with instruments located in engine spaces.) The detent wheel and the signal contacts shall insure proper operation (signal contacts shall close and remain closed for a sufficient period of time) of the audible signals when the transmitter operating knob is moved rapidly from one position to another.

3.5.1.4.6 Signal push switch. - In addition to the signal contacts specified in 3.5.1.4.5, this instrument shall be provided with a watertight manually operated push switch for the audible signal circuit emanating from the instrument. The push switch shall be electrically connected in the circuit in parallel with the signal contacts referred to herein, in order to permit "at will" signaling over the audible signal circuits. The signal push switch shall be connected across terminals M/T and M10.

3.5.1.4.7 Terminal markings. - Terminals marked M/T, MM/T, M1, M2, M3, M4, M5, M6, M7, M8, M9, M10, M/I, MM/I, M11, M12, M13, M14, M15, M16, M17, M18, M19 and M20 shall be provided on the terminal strips for the propeller order circuits.

3.5.1.4.7.1 Terminals for audible signal. - Terminals MM and M20 shall be used to provide connections for an audible signal. The audible signal shall be provided by a type IC/Z1S4 buzzer located on the lower section of the console (see item (d) of 3.5.1.1.2). The type IC/Z1S4 buzzer will be energized by means of a signal switch in the engine room indicator-transmitter.

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3.5.1.5 Enclosure. - The upper or drum section of this console shall be of cast aluminum in accordance with 3.2.2.8.1. The lower section of this console shall be in accordance with 3.3.1.2.1.

3.5.1.6 Mounting. - Sufficient mounting holes for attaching the console to the ships' deck shall be provided to preclude the failure of the mounting attachment during high shock conditions.

3.5.2 Type IC/30D02. - This console shall be arranged in general accordance with Drawing S6505-1,382,706. The console shall contain the instruments and controls for transmitting engine orders (port and starboard), for transmitting propeller revolution orders, and for controlling the ship's aircraft warning and speed lights.

3.5.2.1 Console instrumentation. - The console shall consist of two sections, an upper or drum section and a lower section.

3.5.2.1.1 Upper or drum section. - The upper or drum section shall contain the following equipment:

(a) Engine order indicator-transmitter (Circuit 1MB - starboard).
(b) Engine order indicator-transmitter (Circuit ZMB - port).
(c) Pilot light for each of (a) and (b).
(d) Push switch for each of (a) and (b).
(e) One operating handle for each of (a) and (b).
(f) Dimming rheostat for illumination of (a) and (b).
(g) Speed light circuit control switch.
(h) Speed light signal selector switch.
(i) Operating knob for each of (g) and (h).
(j) Indicator lights for speed light system.
(k) Hand pulse push switch for speed light.

3.5.2.1.2 Lower section. - The lower section shall contain the following equipment:

(a) Propeller order indicator-transmitter.
(b) Push switch for (a).
(c) Operating knobs for (a).
(d) Type IC/Z1S4 buzzer for (a).
(e) Type IC/B1S4 bell for each of (a) and (b) of 3.5.2.1.1.
(f) Dimming rheostat for illumination of (a) and also for speed light controls, (g) and (h) of 3.5.2.1.1.

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3.5.2.2 Engine order indicator-transmitter (Circuit 1MB - starboard). - This indicator-transmitter shall provide a means for transmitting starboard engine orders and receiving acknowledgement of these orders between the console and the starboard main engine control station. This indicator-transmitter shall contain the mechanism necessary to transmit engine orders (and indicate the orders transmitted), and to indicate the replies (acknowledgements) received. The transmitter shall be positioned by means of an operating handle pivoted on the axis and at the end of the drum section as shown on Drawing S6505-1, 382, 706. This handle shall contain the pointer indicating on the right hand dial the order transmitted. The acknowledgement of a starboard engine order shall also be indicated on the right hand dial.

3.5.2.2.1 Dial markings. - Dial markings, lettering, and graduations shall be in accordance with figure 1 of Drawing 9000-S6504-73687. Dials shall be 8 inches in diameter in accordance with Specification MIL-I-983. The dial markings shall be white against a black background.

3.5.2.2.2 Pointers. - The pointer showing orders shall be incorporated with the operating handle. The pointer indicating acknowledgement of engine orders shall be located within the dial windows and shall be white.

3.5.2.2.3 Synchros. - A type 23TR6A synchro unit shall be used to position the acknowledgement pointer. A type 5G synchro unit shall be used for transmitting the engine order. Each synchro unit shall be adjustable in position and shall be so adjusted in position that the associated pointer indicates the center of the "STOP" position on the dial when the synchro unit is at electrical zero.

3.5.2.2.4 Operating handle. - The operating handle shall contain a means of retaining position at a given order and shall require a positive action to disengage the handle when it is desired to move it to a new position.

3.5.2.2.5 Detents. - The transmitter generator operating shaft shall be fitted with a detent wheel to insure stability of position and to operate electric contacts in an audible signal circuit. (The audible signal circuit is used to attract attention of engine room personnel to a change in engine orders.) The detent wheel and the signal contacts shall insure proper operation (signal contacts shall close and remain closed for a sufficient period of time) of the operating handle (or knob) as it is moved rapidly from one position to another. (In knob operated transmitters, the detent wheel shall be provided with a stop at a point corresponding to the division line between the bottom segment on the dial and the adjacent blank segment.)

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3.5.2.2.6 Signal push switch. - A push switch in accordance with 3.5.1.2.6 shall be provided for the engine order system. The terminals for connection of this switch shall be as outlined below.

3.5.2.2.7 Pilot light. - A pilot light in accordance with 3.5.1.2.7 shall be provided for engine order system.

3.5.2.2.8 Illumination power supply. - This power supply shall be in accordance with 3.5.1.2.8.

3.5.2.2.9 Illumination control. - This control shall be in accordance with 3.5.1.2.9.

3.5.2.2.10 Terminal markings. - Terminals marked 1MB/T, 1MBB/T, 1MB1, 1MB2, 1MB3, 1MB/I, 1MBB/I, 1MB4, 1MB5, 1MB6, 1MB7, and 1MB8 shall be provided on the terminal strips for the starboard engine order circuit. Terminals marked 1MB/T and 1MB7 shall be used for connection of the signal push switch.

3.5.2.2.10.1 Terminals for audible signal. - Terminals 1MBB/I and 1MB8 shall be used to provide connections for an audible signal. The audible signal shall be provided a type 1C/B1S4 bell located on the lower section of the console (see item (e) of 3.5.2.1.2). The type 1C/B1S4 bell will be energized by means of a signal switch in the engine room indicator-transmitter. The bell will be located as shown on Drawing S6505-1, 382, 706.

3.5.2.3 Engine order indicator-transmitter (circuit 2MB - port). - This indicator-transmitter shall provide a means for transmitting port engine orders and receiving acknowledgement of these orders between the console and the port main engine control station. This indicator-transmitter shall contain the mechanism necessary to transmit engine orders (and indicate the orders transmitted), and to indicate the replies (acknowledgements) received. The transmitter shall be positioned by means of an operating handle pivoted on the axis and at the ends of the drum section as shown on Drawing S6505-1, 382, 706. This handle shall contain the pointer indicating on the left hand dial the order transmitted. The acknowledgement of a port engine order shall be indicated on the left hand dial.

3.5.2.3.1 Dial markings. - Dial markings lettering and graduations shall be in accordance with figure 2 of Drawing 9000-S6504-73687. Dials shall be 8 inches in diameter in accordance with Specification MIL-I-983. The dial markings shall be white against a black background.

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3.5.2.3.2 Other features. - Other features of this indicator-transmitter shall be in accordance with 3.5.2.2.2, 3.5.2.2.3, 3.5.2.2.4, 3.5.2.2.5, 3.5.2.2.6, 3.5.2.2.7, 3.5.2.2.8, and 3.5.2.2.9.

3.5.2.3.3 Terminal markings. - Terminals marked 2MB/T, 2MBB/T, 2MB1, 2MB2, 2MB3, 2MB/I, 2MBB/I, 2MB4, 2MB5, 2MB6, 2MB7, and 2MB8 shall be provided on the terminal strips for the port engine order circuit. Terminals for the signal push switch shall be 2MB/T and 2MB7.

3.5.2.3.3.1 Terminals for audible signal. - Terminals 2MBB/I and 2MB8 shall be used to provide connections for an audible signal. The audible signal shall be provided by a type 1C/B1S4 bell located on the lower section of the console (see item (e) of 3.5.2.1.2). The type 1C/B1S4 bell will be energized by means of a signal switch in the engine room indicator-transmitter. The bell will be located as shown on Drawing S6505-1,382,706.

3.5.2.4 Speed light controller. - The speed light controller shall be in accordance with 3.5.1.3.

3.5.2.5 Propeller order indicator-transmitter. - This indicator-transmitter shall be in accordance with 3.5.1.4.

3.5.2.6 Other features of console. - Other features which apply to this console shall be in accordance with 3.5.1.5 and 3.5.1.6.

3.5.3 Type IC/30D02. - This console shall be in accordance with 3.5.2, except type 6G synchro units shall be used for the port and starboard engine order transmitters in lieu of type 5G synchro units.

3.5.4 Type 1C/32D04. - This console shall be arranged in general accordance with Drawing S6505-1,383,195. The console shall contain the instruments and controls for transmitting engine orders, and for control of the engine speed and propeller pitch.

3.5.4.1 Instrument panel. - The instrument panel (or top) shall consist of two sections, an upper section at a slope of 45 degrees with the horizontal and a lower section at a slope of 30 degrees with the horizontal.

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3.5.4.1.1 Upper section. - The upper section shall contain the following equipment:

(a) Engine order indicator-transmitter (port).
(b) Engine order indicator-transmitter (starboard).
(c) One pilot light for each of (a) and (b).
(d) One signal push switch for each of (a) and (b).
(e) One operating knob for each of (a) and (b).
(f) Propeller revolution indicator (port).
(g) Propeller revolution indicator (starboard).
(h) Two clutch position lights associated with each of (f) and (g).
(i) Dial illumination control.
(j) Propulsion control order switch.
(k) Propulsion control order indicator lights.

3.5.4.1.2 Lower section. - The lower section shall contain the following equipment:

(a) Propeller pitch controller and indicator (port).
(b) Propeller pitch controller and indicator (starboard).
(c) Throttle controllers (port).
(d) Throttle controllers (starboard).
(e) Engine control air pressure gage.
(f) One pilot light for each of (a) and (b).
(g) Dial illumination control.

3.5.4.2 Engine order indicator-transmitter (Circuit 1MB - starboard). - This indicator-transmitter shall provide a means for transmitting starboard engine orders and receiving acknowledgement of these orders between the console and the starboard main engine control station. This indicator-transmitter shall contain the mechanism necessary to transmit engine orders (and indicate the orders transmitted), and to indicate the replies (acknowledgements) received. The transmitter shall be positioned by means of an operating knob located below the instrument window as shown on Drawing S6505-1, 383, 195. The orders and acknowledgements of a starboard engine order shall be indicated on the right hand instrument.

3.5.4.2.1 Dial markings. - Dial markings, lettering, and graduations shall be in accordance with figure 1 of Drawing 9000-S6504-73687. Dials shall be 4 inches in diameter in accordance with Specification MIL-I-983. The dial markings shall be white against a black background.

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3.5.4.2.2 Pointers.- Two concentric pointers shall be provided. The color of the inner pointer shall be red with a distinctive black and white border, and shall indicate the transmitted order. The outer pointer shall be white, and shall indicate the reply (acknowledgement) to the transmitted order. The inner pointer shall be engraved, or otherwise marked "⊙". The outer pointer shall be engraved, or otherwise marked "A". The marking for each pointer shall be black.

3.5.4.2.3 Synchros.- A type 23TR6A synchro unit shall be used for positioning the outer pointer. A type 5G synchro unit shall be used for transmitting the engine order, and shall be positioned in association with the inner pointer. Each synchro unit shall be adjustable in position, and shall be so adjusted in position that the associated pointer indicates the center of the "STOP" position on the dial when the synchro unit is at electrical zero.

3.5.4.2.4 Operating knob.- The type 5G synchro unit and the associated (inner) pointer shall be positioned by means of an operating knob. Clockwise rotation of the operating knob shall result in clockwise rotation of the pointer. The provisions of 3.3.1.10, 3.3.1.11, and 3.3.1.12 shall apply.

3.5.4.2.5 Other features.- Other features of this indicator-transmitter shall be in accordance with 3.5.2.2.5, 3.5.2.2.6, 3.5.2.2.7, 3.5.2.2.8, 3.5.2.2.9, 3.5.2.2.10, and 3.5.2.2.10.1.

3.5.4.3 Engine order indicator-transmitter (Circuit 2MB - port).- This indicator-transmitter shall provide a means for transmitting port engine orders and receiving acknowledgements of these orders between the console and the port main engine control station. This indicator-transmitter shall contain the mechanism necessary to transmit engine orders (and indicate the orders transmitted), and to indicate the replies (acknowledgements) received. The transmitter shall be positioned by means of an operating knob located below the instrument window as shown on Drawing S6505-1,383,195. The orders and acknowledgements of a port engine order shall be indicated on the right hand instrument.

3.5.4.3.1 Other features.- Other features of this indicator-transmitter shall be in accordance with 3.5.2.2.5, 3.5.2.2.6, 3.5.2.2.7, 3.5.2.2.8, 3.5.2.2.9, 3.5.2.3.3, 3.5.2.3.3.1, 3.5.4.2.1, 3.5.4.2.2, 3.5.4.2.3 and 3.5.4.2.4.

3.5.4.4 Propeller revolution indicators (starboard and port).- These indicators will be furnished by the Government for mounting on the ship control console.

3.5.4.4.1 Mounting.- The opening for console mounting of the propeller revolution indicator will be supplied by the Bureau of Ships upon request.

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3.5.4.4.2 Terminal markings.- Terminals marked 1K, 1KK, 1K4, + (plus) and - (minus) shall be provided for the starboard propeller revolution indicator. Terminals marked 2K, 2KK, 2K4, + (plus) and - (minus) shall be provided for the port propeller revolution indicator.

3.5.4.4.3 Illumination.- Illumination with dimming control will be incorporated in the Government furnished indicator. The power input for illumination shall be 115 volts since an illumination transformer is included with the indicator.

3.5.4.5 Clutch position indicators.- This system is comprised of two indicator lights to indicate whether the clutch is engaged or disengaged. The indicator lights shall be marked "IN" and "OUT" and shall be red in accordance with Specification MIL-I-983. Two indicator lights shall be provided for each shaft and shall be located immediately above their respective propeller revolution indicator as shown on Drawing S6505-1, 383, 195.

3.5.4.5.1 Terminal markings.- Terminals marked 1PWW, 1PW1, and 1PW2 shall be provided for the starboard clutch position indicator lights. Terminals marked 2PWW, 2PW1, and 2PW2 shall be provided for the port clutch position indicator lights.

3.5.4.6 Propulsion control order switch and indicators.- This system is used to transmit orders to the engine room to either assume or relinquish propulsion control and to indicate that the orders have been received. The switch shall be a type JR-304 switch in accordance with Drawing 9000-S6202-73193. The switch shall have two positions marked "Engine room control" and "Pilot house control". Two indicator lights marked "P.H." and "E.R." and of a red color in accordance with Specification MIL-I-983, shall be mounted adjacent to the switch. Two of these lights shall be provided for indicating location of control for both the starboard and port shafts.

3.5.4.6.1 Terminal markings.- Terminals marked 4MB10, 4MBB10, 4MB11, 4MB12, 4MB13 and 4MB14 shall be provided for the starboard propulsion control order system. Terminals marked 4MB20, 4MBB20, 4MB21, 4MB22, 4MB23 and 4MB24 shall be provided for the port propulsion control order system. Wiring connections between the equipment components and the terminal strips will be supplied by the Bureau of Ships on request.

3.5.4.6.2 Other features.- Provisions shall be made to indicate the position of the switch when the pilot house is operating under darkened conditions.

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3.5.4.7 Dial illumination (upper section). - One master dimming rheostat will be provided for the upper section of the console to control the intensity of illumination for the engine order indicator-transmitters. Separate rheostat sections shall be provided for each instrument. This control shall be provided with an "OFF" position.

3.5.4.8 Propeller pitch controller and indicator (starboard and port). - These controllers will be furnished by the Government for mounting on the ship control console. They will be arranged on the console in general accordance with Drawing S6505-1,383,195. The opening for console mounting shall be as shown on the sketch included with Specification MIL-C-16238, except no provisions for dowel holes are required.

3.5.4.8.1 Pilot lights. - Pilot lights in accordance with 3.5.1.2.7 shall be provided for each controller and indicator.

3.5.4.8.2 Terminals. - Thirty-six (36) terminals shall be provided for the starboard and port propeller pitch controller and indicators. The markings of these terminals will be furnished by the Government upon request.

3.5.4.9 Throttle controllers (starboard and port). - The starboard and port throttle controllers will be furnished by the ship contractor. Plans of this equipment will be furnished by the Bureau of Ships to aid the manufacturer in determining the mounting template.

3.5.4.9.1 Union fittings. - Provisions shall be made for connections by means of union fittings to a 1/2 (one-half) inch actual outside diameter copper tube containing 100 pounds per square inch air supply; to one 3/8 (three-eighths) inch actual outside diameter copper tube containing controlled air for the starboard throttle control; and to one 3/8 (three-eighths) inch actual outside diameter copper tube containing controlled air for the port throttle control. These three union fittings will be brought approximately to deck level on the inside of the console. These union fittings for connection to ships' pneumatic piping shall be in accordance with the straight union fitting of Specification MIL-F-1183.

3.5.4.9.2 Air pressure gage. - The air pressure gage shall be mounted on the console as shown on Drawing S6505-1,383,195. Provisions for mounting dimensions and dial markings shall be as shown on Drawing S4105-3,119,768 (dial "A" shall be used). Piping connections shall be as shown on figure 1.

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3.5.4.9.3 Connection tubing.- The manufacturer shall make internal connections in the console from the 1/2 (one-half) inch diameter air supply line to the starboard throttle controller (using 3/8 (three-eighths) inch actual outside diameter copper tubing with 1/4 (one-fourth) inch I.P.S. in accordance with Specification WW-T-799), to the port throttle controller (using 3/8 (three-eighths) inch actual outside diameter tubing with 1/4 (one-fourth) inch I.P.S. in accordance with Specification WW-T-799); and to the pressure gage (using 3/8 (three-eighths) inch actual outside diameter tubing with 1/4 (one-fourth) inch I.P.S. in accordance with Specification WW-T-799). The fittings shall be of the silver-brazed type equivalent to those specified in Specification MIL-F-1183. Tubing shall be as shown on figure 1.

3.5.4.10 Dial illumination (lower section).- One master dimming rheostat will be provided for the lower section of the console to control the intensity of illumination for the propeller pitch controller and indicators (starboard and port) and the air pressure gage. Separate rheostat sections shall be provided for each instrument. This control shall be supplied with an "OFF" position.

3.5.4.11 Upper panel cover.- The upper section of the console shall be a panel cover in accordance with 3.3.1.14.

3.5.4.12 Mounting.- Mounting shall be in accordance with 3.5.1.6.

3.6 Steering control consoles.-

3.6.1 Type IC/50D01.- This console shall be arranged in general accordance with Drawing S6505-1,382,705. It shall contain the instruments necessary for steering the ship.

3.6.1.1 The instrument panel (or top) shall contain the following equipment:

(a) Rudder angle-order indicator-transmitter (circuits L and N).
(b) Rudder order transmitter signal push switch.
(c) Rudder order transmitter pilot light.
(d) Magnesyn compass indicator (circuit LM).
(e) Ship's course indicator (circuit LC).
(f) Steering emergency switch (circuit LB).
(g) Helm angle indicator.
(h) Master dimming rheostat.

3.6.1.2 The front of the console shall contain the following equipment:

(a) Steering wheel and helm angle transmitter.

3.6.1.3 The right hand side of the console shall contain the following equipment:

(a) Cable selector switch.

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3.6.1.4 Rudder order-angle indicator-transmitter (circuits L and N). - This instrument shall provide a means for transmitting and indicating the desired rudder angle, and for receiving and indicating rudder angle signals transmitted by a rudder angle transmitter located at the ship's rudder.

3.6.1.4.1 Dial markings. - Dial markings, lettering, and graduations shall be in accordance with figure 7 of Drawing 9000-S6504-73687.

3.6.1.4.2 Pointers. - The outer pointer, positioned by a type 23TR6A synchro unit, shall be engraved or otherwise marked, "ANGLE". The outer pointer shall be white with black markings. The inner pointer, positioned in conjunction with a type 5G synchro unit by the operating knob, shall be engraved, or otherwise marked, "ORDER". The inner pointer shall be red with black markings and with a distinctive black and white border.

3.6.1.4.3 Positioning of pointers. - The outer pointer shall be positioned by a type 23TR6A synchro unit. The inner pointer shall be positioned in conjunction with a type 5G synchro unit by means of an operating knob located adjacent to the instrument dial. The provisions of 3.3.6 shall apply.

3.6.1.4.4 Synchros. - Each synchro unit shall be adjustable in position and shall be so adjusted in position that the associated pointer indicates the "O" position on the dial when the synchro unit is at electrical zero.

3.6.1.4.5 Rudder order push switch. - A watertight manually operated push switch in accordance with Drawing 9000-S6202-73754 shall be provided for the audible signal circuit emanating from the console section in association with the rudder order system.

3.6.1.4.6 Pilot light. - The pilot light shall be in accordance with 3.5.1.2.7.

3.6.1.4.7 Illumination. - Illumination for this instrument shall be energized from the power supply to the rudder angle indicator (circuit N).

3.6.1.4.8 Terminal markings. - Terminals marked L, LL, L1, L2, L3 and L4 shall be supplied for the rudder order transmitter. Terminals marked N, NN, N1, N2, and N3 shall be supplied for the rudder angle indicator. Terminals L and L4 shall be used for signal push switch circuit connections.

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3.6.1.5 Magnesyn compass indicator - (circuit LM). - This instrument shall provide a means for indicating ship's course as determined by the magnetic compass. Two repeaters (the second repeater to be included in the maintenance parts kit) will be furnished by the Government. Both instruments shall be equipped by the contractor with a means of illumination which is in general accordance with 3.4.9.

3.6.1.5.1 Terminal markings. - Terminals marked LM-A, LM-B, LM-C and LM-D shall be supplied on the terminal strips for the magnesyn compass system.

3.6.1.6 Ship's course indicator - (circuit LC). - This indicator will be supplied by the Government for mounting in the console. Illumination will be built into the indicator and terminals will be provided for connection to a dimming control system. Plans for mounting dimensions of the indicator will be supplied by the Government.

3.6.1.6.1 Terminal markings. - Terminals marked LC, LCC, LC1, LC2, LC3, LC4, LC5, and LC6 shall be provided on the terminal strips of the console for the ship's course indicator.

3.6.1.7 Steering emergency alarm switch - (circuit LB). - This switch shall incorporate a spring return type S-2JRM3-22L switch in accordance with Drawing 9000-S6504-74055. The switch shall have a positive lock when in the "OFF" position. The switch shall be provided with an operating lever approximately 2-1/2 inches in length. The operating lever shall rotate clockwise from the "OFF" position to the "ON" position. The "OFF" of the lever shall be at the 3 o'clock position, and the "ON" position shall be at 4:30 o'clock. Wiring of the switch shall be as shown on figure 3 of Drawing 9000-S6504-74055. The switch shall be located in general accordance with Drawing S6505-1,382,705-Rev. 1.

3.6.1.7.1 Terminal markings. - Terminals marked LB and LBB shall be provided on the terminal strip of the console for the steering emergency switch.

3.6.1.8 Helm angle indicator. - This instrument shall provide a means of indicating the helm angle as set by the helm angle transmitter (item (a) of 3.6.1.2). A direct mechanical coupling shall be provided between the indicator and the Government-furnished helm angle transmitter. Illumination shall be provided in accordance with 3.4.9. The dial of this indicator shall be 4 inches in diameter in accordance with Specification MIL-I-983. The precise dial markings for the helm angle indicator will be supplied by the Government upon request. The pointer of the instrument shall be white.

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3.6.1.9 Helm angle transmitter. - This instrument will be furnished by the Goverment for mounting on the front of the consc'e. The steering wheel provided with the transmitter will be 12 inches in diameter. Plans for mounting dimensions of the transmitter will be supplied by the Government upon request.

3.6.1.9.1 Terminal markings. - Terminals marked LCS1, LCS2, LCS3, LCS4, LCS5, and LCS6 shall be provided on the terminal strips of the console for the port cable to the helm angle transmitter, and terminals marked LCS11, LCS12, LCS13, LCS14, LCS15, and LCS16 shall be provided for the starboard cable.

3.6.1.10 Cable selector switch. - This switch shall provide a means of transferring the helm angle transmitter to either the port or starboard cable. This switch will be furnished by the Government with the helm angle transmitter. The switch shall be set into the right side panel of the console, and located in such manner as to prevent any portion of the switch from protruding beyond the surface of the right side panel. The switch will be a type S-2JR switch in accordance with Drawing 9000-S6202-73826. The switch shall be located in general accordance with Drawing S6505-1, 382, 705.

3.6.1.11 Illumination control. - One master dimming rheostat with separate sections for all components requiring illumination shall be provided on the instrument panel of the console for control of the intensity of illumination. This control will be provided with an "OFF" position.

3.6.1.12 Panel cover. - The instrument cover panel shall be in accordance with 3.3.1.14.

3.6.1.13 Mounting. - The mounting of the console shall be in accordance with 3.5.1.6.

3.6.1.14 Special provisions of type IC/50D01 console. - In the interest of basic design standardization for the steering control consoles covered by this specification, the space required by an after-gyro ship's course indicator, or a course-to-steer indicator, or a course-steering double indicator (symbol 2 of Drawing S6505-1, 382, 705) shall be provided in the type 1C/50D01 steering control console. Structural members, as needed for eventual mounting of such an additional instrument, shall be provided. The dimensions shall be equivalent to those of the indicator outlined in 3.6.1.6.

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3.6.2 Type 1C/51D02.- This console shall be arranged in general accordance with Drawing S6505-1, 382, 705. It shall contain the instruments necessary for steering the ship. The provisions of 3.6.1.2, 3.6.1.3, 3.6.1.4, 3.6.1.5, 3.6.1.6, 3.6.1.7, 3.6.1.8, 3.6.1.9, 3.6.1.10, 3.6.1.11, 3.6.1.12, and 3.6.1.13 shall apply.

3.6.2.1 The instrument panel (or top) shall contain the following equipment:

(a) Rudder angle-order indicator-transmitter (circuits L and N).
(b) Rudder order transmitter signal push switch.
(c) Rudder order transmitter pilot light.
(d) Magnesyn compass indicator (circuit LM).
(e) Ship's course indicator (circuit LC).
(f) Steering emergency switch (circuit LB).
(g) Helm angle indicator.
(h) Master dimming rheostat.
(i) Course-steering order indicator (circuit LA).
(j) Course-steering order power switch.
(k) Course-steering order indicator light.
(ℓ) Course-steering order acknowledgement push switch.

3.6.2.2 Course-steering order indicator.- This instrument shall provide a means for receiving and indicating course and steering orders emanating from a course-steering transmitter located on the open bridge.

3.6.2.2.1 Dial markings.- Dial markings, lettering, and graduations shall be in accordance with figure 10 of Drawing 9000-S6504-73687. The inner portion of the dial, marked for steering orders and including the fixed reference point above "CORS", shall be fixed. The outer portion of the dial, marked for course orders, rotates about the fixed portion of the dial.

3.6.2.2.2 Pointer.- A single pointer shall be provided for indication of the steering order. The pointer shall be positioned directly by a type 23TR6A synchro receiver. The pointer shall be white.

3.6.2.2.3 Course dial.- Indication of the course order shall be provided by means of rotation of the course dial ring, relative to the fixed indication mark shown above "CORS" on figure 10 of Drawing 9000-S6504-73687. The course dial ring shall be positioned by a type 5F synchro receiver. The course dial ring shall be geared to the associated synchro receiver, and the motion transferred between the synchro receiver gear and the course dial gear shall be in a 1 to 1 ratio.

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3.6.2.2.4 Audible signal. - A type IC/B1S4 bell shall be provided in the console for attention signals from a course-steering order transmitter on the open bridge. This bell shall be accessible from the front access plate of the console.

3.6.2.2.5 Indicator light. - An indicator light, connected in parallel with the audible signal, shall be provided on the instrument panel of console for visual indication of a change in course-steering orders. This indicator light shall be of a red color in accordance with Specification MIL-I-983.

3.6.2.2.6 Power switch. - A switch shall be provided on the instrument panel of the console for controlling the power to the course-steering order system. The switch shall be connected between the rudder angle indicator power leads (N and NN) and the power leads for the course-steering system (LA/C, LAA/C, LA/S and LAA/S). The switch leads on the rudder angle side shall be fused. The switch shall be provided with "OFF" and "ON" positions, and shall be a type JR-304 switch in accordance with Drawing 9000-S6202-73193.

3.6.2.2.7 Acknowledgement push switch. - A push switch in accordance with Drawing 9000-S6202-73754 shall be provided on instrument panel of the console to acknowledge to the course-steering transmitter station that a particular order has been received.

3.6.2.2.8 Terminal markings. - Terminals marked LA/C, LAA/C, LA1, LA2, LA3, shall be provided for connection to the type 5F synchro unit associated with the course order indication. Terminals marked LA/S, LAA/S, LA4, LA5, and LA6 shall be provided for connection to the type 23TR6 synchro receiver. Terminals marked LA7 and LA8 shall be provided for connection to the signal pushbutton and audible signal, respectively. The signal push switch incorporated in this indicator shall be connected across terminals LA and LA7. The audible signal shall be connected across terminals LAA and LA8.

3.6.3 Type 1C/52D01. - This console shall be arranged in general accordance with Drawing S6505-1, 382, 705. It shall contain the instruments necessary for steering the ship. The provisions of 3.6.1.3, 3.6.1.4, 3.6.1.5, 3.6.1.6, 3.6.1.7, 3.6.1.8, 3.6.1.10, 3.6.1.11, 3.6.1.12, and 3.6.1.13 shall apply.

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3.6.3.1 The instrument panel (or top) shall contain the following equipment:

(a) Rudder angle-order indicator-transmitter (circuit L and N).
(b) Rudder order transmitter signal push switch.
(c) Rudder order transmitter pilot light.
(d) Magnesyn compass indicator (circuit LM).
(e) Ship's course indicator (forward gyro-circuit LC).
(f) Ship's course indicator (after gyro-circuit LC-A).
(g) Helm angle indicator.
(h) Steering emergency switch (circuit LB).
(i) Master dimming rheostat.

3.6.3.2 The front of the console shall contain the following equipment:

(a) Steering wheel and helm angle transmitter.

3.6.3.3 The right hand side of the console shall contain the following equipment:

(a) Cable selector switch.

3.6.3.4 Ship's course indicator (after gyro-circuit LC-A).- This indicator will be supplied by the Government for mounting on the console. Illumination will be built into the indicator and terminals will be provided for connection to a dimming control system. Plans for mounting dimensions of the indicator will be supplied by the Government.

3.6.3.4.1 Terminal markings.- Terminals marked LCA, LCCA, LCA1, LCA2, LCA3, LCA4, LCA5, and LCA6 shall be provided on the terminal strips of the console for the after gyro ship's course indicator.

3.6.4 Type IC/53D01.- This console shall be arranged in general accordance with Drawing S6505-1,382,705. It shall contain the instruments necessary for steering the ship. The provisions of 3.6.1.2, 3.6.1.5, 3.6.1.6, 3.6.1.7, 3.6.1.8, 3.6.1.9, 3.6.1.11, 3.6.1.12, 3.6.1.13, and 3.6.3.4 shall apply.

3.6.4.1 The instrument panel (or top) shall contain the following equipment:

(a) Rudder angle-order indicator-transmitter (circuits L, 1N, 2N, and 3N).
(b) Rudder order transmitter signal push switch.
(c) Rudder order transmitter pilot light.
(d) Magnesyn compass indicator (circuit LM).
(e) Ship's course indicator (forward gyro-circuit LC).

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(f) Ship's course indicator (after gyro-circuit LC-A).
(g) Helm angle indicator.
(h) Steering emergency switch (circuit LB).
(i) Master dimming rheostat.

3.6.4.2 The right hand side of the console shall contain the following equipment:

(a) Rudder angle transfer switch.

3.6.4.3 Rudder order-angle indicator-transmitter (circuits L, 1N, 2N, and 3N). This instrument shall provide a means for transmitting and indicating the desired rudder angle, and for receiving and indicating rudder angle signals transmitted by rudder angle transmitters located at the ship's three rudder heads. - The diameter of the indicator-transmitter dial shall be 6 inches.

3.6.4.3.1 Dial markings. - The provisions of 3.6.1.4.1 shall apply.

3.6.4.3.2 Pointers. - The top (or outer) pointer, positioned by a type 23TR6A synchro unit, shall be engraved, or otherwise marked, "P". The next (or second from the top) pointer, positioned by a type 3F synchro unit, shall be engraved, or otherwise marked, "C". The next (or third from the top) pointer, positioned by a type 3F synchro unit, shall be engraved, or otherwise marked, "S". The inner pointer, positioned in conjunction with a type 5G synchro transmitter, shall be engraved, or otherwise marked, "O". The outer (or top pointer) shall be red with black letter markings. The next (or second from the top) pointer shall be white with a distinctive black and white border and with black letter markings. The next (or third from the top) pointer shall be green with a distinctive black and white border and with white letter markings. The inner pointer shall be crosshatched white against black, and with black letter markings (but the center of "O" a solid white).

3.6.4.3.3 Positioning of pointers. - The outer pointer shall be positioned directly by a type 23TR6A synchro unit. The middle pointers shall be positioned by type 3F synchro units. The inner pointer shall be positioned in conjunction with a type 5G synchro unit by means of an operating knob located adjacent to the instrument dial. The provisions of 3.3.6 shall apply.

3.6.4.3.4 Synchros. - Each synchro unit shall be adjustable in position and shall be so adjusted in position that the associated pointer indicates the "O" position on the dial when the synchro unit is at electrical zero.

3.6.4.3.5 Rudder order push switch. - The provisions of 3.6.1.4.5 shall apply.

3.6.4.3.6 Pilot light. - The provisions of 3.5.1.2.7 shall apply.

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3.6.4.3.7 Illumination.- The provisions of 3.6.1.4.7 shall apply.

3.6.4.3.8 Terminal markings.- Terminals marked L, LL, L1, L2, L3, and L4 shall be supplied for the rudder order transmitter. Terminals marked 1N, 1NN, 1N1, 1N2, and 1N3 shall be supplied for the starboard rudder angle indicator. Terminals marked 2N, 2NN, 2N1, 2N2, and 2N3 shall be supplied for the port rudder angle indicator. Terminals marked 3N, 3NN, 3N1, 3N2, and 3N3 shall be provided for the center rudder angle indicator. Terminals L and L4 shall be used for audible signal circuit connections.

3.6.4.4 Rudder angle transfer switch.- This switch shall provide a means of transferring the rudder angle indicators from normal rudder angle transmitters to the emergency rudder angle transmitters. The switch shall be set into the side of console to prevent any portion of the switch from protruding beyond the surface. The switch shall be a type S-2JR switch in accordance with Drawing 9000-S6202-73826. The switch shall have positions marked "Normal", "Emergency", and "OFF". It shall be located in general accordance with the position of the cable selector switch as shown on Drawing S6505-1, 382, 705. The switch shall be provided with a means of illumination to indicate the position of the switch under darkened conditions. This illumination shall be of a red color in accordance with Specification MIL-I-983.

3.6.4.5 Terminal markings.- Terminals marked 4N, 4NN, 4N1, 4N2, and 4N3 shall be provided for the starboard emergency rudder angle circuit. Terminals marked 5N, 5NN, 5N1, 5N2, and 5N3 shall be provided for the port emergency rudder angle circuit. Terminals marked 6N, 6NN, 6N1, 6N2, and 6N3 shall be provided for the center emergency rudder angle system. The terminals for the normal rudder angle indicators shall be as specified in 3.6.4.3.8.

3.6.5 Type IC/54D01.- This console shall be arranged in general accordance with Drawing S6505-1, 382, 705 - Rev. 1. It shall contain the instruments necessary for steering the ship. The provisions of 3.6.1.2, 3.6.1.3, 3.6.1.4, 3.6.1.5, 3.6.1.6, 3.6.1.7, 3.6.1.8, 3.6.1.9, 3.6.1.10, 3.6.1.11, 3.6.1.12, and 3.6.1.13 shall apply.

3.6.5.1 The instrument panel (or top) of the console shall contain the following equipment:

(a) Rudder angle-order indicator-transmitter (circuits L and N).
(b) Rudder order transmitter signal push switch.
(c) Rudder order transmitter pilot light.
(d) Magnesyn compass indicator (circuit LM).
(e) Ship's course indicator (circuit LC).

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(f) Course-to-steer indicator (circuit 2GR).
(g) Steering emergency switch (circuit LB).
(h) Helm angle indicator.
(i) Master dimming switch.

3.6.5.2 Course-to-steer indicator - (circuit 2GR). - This indicator will be supplied by the Government for mounting in the console. Illumination will be built into the indicator and terminals will be provided for connection to a dimming control system. Plans for mounting dimensions of the indicator will be supplied by the Government.

3.6.5.2.1 Terminal markings. - Terminals marked 2GR, 2GRR, 2GR1, 2GR2, 2GR3, 2GR4, 2GR5, and 2GR6 shall be provided on the terminal strips of the console for the course-to-steer indicator.

3.7 Ship control combination instruments-indicators. -

3.7.1 Type IC/12B02. - This instrument shall be arranged in general accordance with Drawing S6505-1,412,982. The instrument shall contain indications of the main ship control functions.

3.7.1.1 The instrument panel shall contain the following equipment:

(a) Engine order indicator - (starboard orders and answers - circuit 1MB).
(b) Engine order indicator - (port orders and answers - circuit 2MB).
(c) Rudder angle indicator - (circuit N).
(d) Propeller revolution indicator - (starboard shafts - circuits 1K and 2K).
(e) Propeller revolution indicator - (port shafts - circuits 3K and 4K).
(f) Underwater log indicator - (circuit Y).
(g) Backing lights associated with (d).
(h) Backing lights associated with (e).

3.7.1.2 Engine order indicator - (starboard orders and answers). - This instrument shall provide a means of indicating the starboard engine orders transmitted from the pilothouse and the answers (or acknowledgement) to these orders. The dial shall be 4 inches in diameter.

3.7.1.2.1 Dial markings. - Dial markings, lettering, and graduations shall be in accordance with figure 1 of Drawing 9000-S6504-73687.

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3.7.1.2.2 Pointers.- The outer pointer shall be engraved, or otherwise marked, "A". The inner pointer shall be engraved, or otherwise marked, "O". The markings, "A" and "O", shall be read from top to bottom when the pointer indicates the center of the "STOP" position on the dial.

3.7.1.2.3 Positioning of pointers.- The outer pointer shall be positioned by a type 23TR6A synchro unit. The inner pointer shall be positioned by means of a type 3F synchro unit. Each synchro shall be adjustable in position and shall be so adjusted in position that the associated pointer indicates the center of the "STOP" position on the dial when the synchro is at electrical zero.

3.7.1.2.4 Illumination.- Illumination shall be provided for this instrument in accordance with 3.4.9. The illumination circuit shall receive its power from the power leads of the indicator involved.

3.7.1.2.5 Terminal markings.- Terminals marked 1MB0, 1MBB0, 1MB1, 1MB2, and 1MB3 shall be provided for the order side of the indicator. Terminals marked 1MBA, 1MBBA, 1MB4, 1MB5, and 1MB6, shall be provided for the answer side of the starboard engine order indicator. Terminals 1MBB0 and 1MB7 shall be provided for audible signal connections.

3.7.1.3 Engine order indicator - (port orders and answers).- This instrument shall provide a means of indicating the port engine orders transmitted from the pilothouse and the answers (or acknowledgement) to these orders. The dial shall be 4 inches in diameter. The provisions of 3.7.1.2.1, 3.7.1.2.2, and 3.7.1.2.3 shall apply.

3.7.1.3.1 Terminal markings.- Terminals marked 2MB0, 2MBB0, 2MB1, 2MB2, and 2MB3 shall be provided for the order side of the port indicator. Terminals marked 2MBA, 2MBBA, 2MB4, 2MB5, and 2MB6 shall be provided for the answer side of the port engine order indicator. Terminals 2MBB0 and 2MB7 shall be provided for the audible signal connection.

3.7.1.4 Rudder angle indicator.- This instrument shall provide a means of indicating the rudder angle. The dial of this instrument shall be 4 inches in diameter. The provisions of 3.7.1.2.3 shall apply.

3.7.1.4.1 Dial marking.- Dial markings, lettering, and graduations shall be in accordance with figure 7 of Drawing 9000-S6504-73687.

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3.7.1.4.2 Pointer.- The pointer shall be engraved, or otherwise marked, "ANGLE". The marking, "ANGLE", shall be read from top to bottom when the pointer indicates the zero position on the dial. The pointer will position by means of a type 23TR6A synchro unit. The synchro shall be adjustable in position, and shall be so adjusted in position that the pointer indicates the "O" position on the dial when the synchro is at electrical zero.

3.7.1.4.3 Terminal markings.- Terminals marked N, NN, N1, N2, and N3 shall be provided on the terminal strip for the rudder angle indicator.

3.7.1.5 Propeller revolution indicator (starboard shafts).- This indicator shall provide a means of indicating the propeller revolutions per minute for the starboard shafts. The dial shall be 4 inches in diameter. The provisions of 3.7.1.2.3 shall apply.

3.7.1.5.1 Dial markings.- Dial markings, lettering, and graduations shall be in accordance with figure 22 of Drawing 9000-S6504-73687.

3.7.1.5.2 Pointers.- The outer pointer shall be engraved, or otherwise marked, "1". The inner pointer shall be engraved, or otherwise marked, "2". The markings, "1" and "2", shall be read from top to bottom when the pointer indicates the "O" position on the dial.

3.7.1.5.3 Positioning of pointers.- The outer pointer shall be positioned by a type 23TR6A synchro unit. The inner pointer shall be positioned by means of a type 3F synchro unit. Each synchro shall be adjustable in position and shall be so adjusted in position that the associated pointer indicates the "O" position when the synchro is at electrical zero.

3.7.1.5.4 Backing signal.- Backing signals shall be provided for this instrument to indicate backing of each shaft. The backing signals shall be located on the instrument panel cover external to the propeller revolution indicator. The signal shall consist of a light visible through an aperture in the instrument panel cover. The color of the light showing shall be red in accordance with Specification MIL-I-983.

3.7.1.5.5 Terminal markings.- Terminals marked 1K, 1KK, 1K1, 1K2, and 1K3 shall be provided for the (outboard) starboard shaft propeller revolution indicator. Terminals marked 2K, 2KK, 2K1, 2K2 and 2K3 shall be provided for the (inboard) starboard shaft propeller revolution indicator. Terminals marked 1K7 and 2K7 shall be provided for the backing lights. The backing light shall be connected across 1K and 1K7 for the (outboard) starboard shaft, and across 2K and 2K7 for the (inboard) starboard shaft.

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3.7.1.6 Propeller revolution indicator - (port shafts). - This indicator shall provide a means of indicating the propeller revolutions per minute of the port shafts. The dial of the instrument shall be 4 inches in diameter. The provisions of 3.7.1.2.3, 3.7.1.5.1, 3.7.1.5.3, and 3.7.1.5.4 shall apply.

3.7.1.6.1 Pointers. - The outer pointer shall be engraved, or otherwise marked, "3". The inner pointer shall be engraved, or otherwise marked, "4". The markings, "3" and "4", shall be read from top to bottom when the pointer indicates the "O" position on the dial.

3.7.1.6.2 Terminal markings. - Terminals marked 3K, 3KK, 3K1, 3K2 and 3K3 shall be provided for the (inboard) port shaft propeller revolution indicator. Terminals marked 4K, 4KK, 4K1, 4K2, and 4K3 shall be provided for the (outboard) port shaft propeller revolution indicator. Terminals marked 3K7 and 4K7 shall be provided for the port backing lights. The backing light shall be connected across 3K and 3K7 for the (inboard) port shaft, and across 4K and 4K7 for the (outboard) port shaft.

3.7.1.7 Underwater log indicator. - This instrument shall provide a means of indicating the ship's speed in knots. The dial of this indicator shall be 4 inches in diameter. The provisions of 3.7.1.2.3 shall apply.

3.7.1.7.1 Dial marking. - Dial markings, lettering, and graduations shall be in accordance with figure 17 of Drawing 9000-S6504-73687.

3.7.1.7.2 Positioning of the pointer. - The pointer shall be positioned by means of a type 23TR6A synchro. The synchro shall be adjustable in position, and shall be so adjusted in position that the associated pointer indicates the "O" position on the dial when the synchro is at electrical zero.

3.7.1.7.3 Terminal markings. - Terminals marked Y, YY, Y1, Y2, and Y3 shall be provided on the terminal strip for the underwater log indicator.

3.7.1.8 Illumination control. - A bank of three rheostats with one operating knob shall be provided in the indicator as shown on Drawing S6505-1,412,982, for control of the instrument illumination. One rheostat shall control the illumination for the engine order indicator-transmitters. A second rheostat shall control the illumination for the underwater log and the rudder angle indicators. A third rheostat shall control the illumination for the propeller revolution indicators. The power for the illumination circuit shall be taken from the secondary of a transformer whose primary is connected across terminals N and NN. The illumination control shall be provided with an "OFF" position.

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3.7.1.9 Condensation drain.- A condensation drain shall be provided on this instrument at the lowest point of the enclosure.

3.7.1.10 Mounting.- This instrument shall be provided with three point bulkhead mounting in accordance with Specification MIL-I-983.

3.7.1.11 Enclosure.- The enclosure shall be cast aluminum in accordance with 3.2.2.8.1.

3.7.1.12 Terminal strip access plate.- The instrument shall be provided with an access plate to facilitate the connection of ship's wiring to the instrument terminal strips. Sufficient space shall be provided at the terminal strip to preclude overcrowding of this space when ship's wiring is installed. The terminal strip access plate shall be furnished with an "O" ring gasket to assure watertight integrity of the installation.

3.7.1.13 Cable entrance.- Provisions shall be made for the entrance of two type DSGA-3 or two type DHFA-3 cables for audible signal circuits for the engine order systems. Provisions shall also be made for entrance of two type MSCA-37 or type MHFA-37 cables for ship's wiring. In addition to the above entrance holes, boss area shall be provided for the entrance of a type MSCA-24 or type MHFA-24 cable at a later date. Entrance holes shall be provided for the smaller diameter cable with sufficient boss area for possible entrance of the larger diameter cable. The provisions of 3.3.4 shall apply.

3.7.1.14 Special provision of types 1C/12B02, and 1C/13B02 ship control indicators.- In the interest of basic design standardization for the ship control indicators of type 1C/12B02 and 1C/13B02 covered by this specification, the enclosure design for the type 1C/13B02 indicator shall be applicable to the type 1C/12B02 indicator.

3.7.2 Type IC/13B02.- This instrument shall be arranged in general accordance with Drawing S6505-1,412,982. The instrument shall contain indications of the main ship control functions. The provisions of 3.7.1.2, 3.7.1.3, 3.7.1.5, 3.7.1.6, 3.7.1.7, 3.7.1.8, 3.7.1.9, 3.7.10, 3.7.1.11, 3.7.1.12, 3.7.1.13, and 3.7.1.14 shall apply.

3.7.2.1 The instrument panel shall contain the following equipment:

(a) Engine order indicator - (Starboard orders and answers - circuit 1MB).
(b) Engine order indicator - (Port orders and answers - circuit 2MB).
(c) Rudder angle indicator - (circuits 1N, 2N, and 3N).
(d) Propeller revolution indicator - (Starboard shafts - circuits 1K and 2K).

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(e) Propeller revolution indicator - (Port shafts - circuits 3K and 4K).
(f) Underwater log indicator - (circuit Y).
(g) Backing lights associated with (d).
(h) Backing lights associated with (e).

3.7.2.2 Rudder angle indicator - (circuits 1N, 2N, and 3N). - This instrument shall provide a means of indicating the angular positions of the rudders. The dial of this instrument shall be 4 inches in diameter. The provisions of 3.7.1.2.3 and 3.7.1.4.1 shall apply.

3.7.2.2.1 Pointers. - The outer pointer, directly positioned by a type 23TR6A synchro unit, shall be engraved, or otherwise marked, "P". The middle pointer, positioned by a type 3F synchro unit, shall be engraved, or otherwise marked, "C". The inner pointer, positioned by means of a type 3F synchro unit, shall be engraved, or otherwise marked, "S". The outer pointer shall be red with black letter markings. The middle pointer shall be white with a distinctive black and white border and with black letter markings. The inner pointer shall be green with a distinctive black and white border and with white letter markings.

3.7.2.2.2 Terminal marking. - Terminals marked 1N, 1NN, 1N1, 1N2, and 1N3 shall be provided for the starboard rudder angle indicator. Terminals marked 2N, 2NN, 2N1, 2N2, and 2N3 shall be provided for the port rudder angle indicator. Terminals marked 3N, 3NN, 3N1, 3N2, and 3N3 shall be provided for the center rudder angle indicator.

3.8 Ship control combination instruments - indicator-transmitters. -

3.8.1 Type IC/14S03. - This indicator-transmitter shall contain the instruments and controls for transmitting engine orders, for transmitting propeller revolution orders and for receiving and indicating the answers to these orders. The provisions of 3.5.1.4, 3.5.2.2, and 3.5.2.3, shall apply.

3.8.1.1 Instrumentation. - The indicator-transmitter shall consist of three sections: an upper or drum section, a middle section, and a lower section or pedestal.

3.8.1.1.1 Upper or drum section. - The upper or drum section shall contain the following:

(a) Engine order indicator-transmitter (circuit 1MB - starboard).
(b) Engine order indicator-transmitter (circuit 2MB - port).
(c) Pilot light for each of (a) and (b).
(d) Push switch for each of (a) and (b).
(e) One operating handle for each of (a) and (b).
(f) Dimming rehostat for each of (a) and (b).

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3.8.1.1.2 Middle section. - The middle section shall contain the following equipment:

(a) Propeller order indicator-transmitter.
(b) Push switch for (a).
(c) Operating knobs for (a).
(d) Type IC/21S4 buzzer for (a).
(e) Type IC/B1S4 bell for each of (a) and (b) of 3.8.1.1.1.
(f) Dimming rheostat for (a).

3.8.1.1.3 Lower section. - The lower section shall consist of a pedestal for mounting the equipment.

3.8.1.2 Size and weight. - The overall dimensions of the indicator-transmitter, including the handles for operating the engine order transmitters, and weight of the indicator-transmitter shall not exceed the following:

Pedestal mounting

Height - 53 inches.
Width - 15 inches.
Depth - 14 inches.
Weight - 190 pounds.

3.8.1.3 Enclosure. - The three sections of this indicator-transmitter shall be cast aluminum in accordance with 3.2.2.8.1.

3.8.1.4 Access plate. - The terminal strip access plate shall be located on the back of the middle section of the indicator-transmitter. Connection of ship's wiring to equipment's wiring will be made at this access plate, after the ship's wiring is brought through the base of the indicator-transmitter.

3.8.1.5 Terminal tube plate. - A terminal tube plate shall be provided between the middle section and the pedestal in accordance with Specification MIL-I-983.

3.8.1.6 Condensation drain. - A condensation drain shall be located in the terminal tube plate, and the outlet of the drain shall be located in the outer (exposed) surface of the terminal tube plate. The condensation drain shall be in accordance with Specification MIL-I-983.

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3.8.2 Type IC/11B01.- This instrument shall be arranged in general accordance with Drawing S6505-1,430,783. The instrument shall contain indicators of the main ship control functions. The provisions of 3.7.1.9, 3.7.1.10, 3.7.1.11, and 3.7.1.12 shall apply.

3.8.2.1 The instrument panel shall contain the following equipment:

(a) Engine order indicator-transmitter (Starboard - circuit 1MB).
(b) Engine order indicator-transmitter (Port - circuit 2MB).
(c) One pilot-light for each of (a) and (b).
(d) One signal push switch for each of (a) and (b).
(e) One operating knob for each of (a) and (b).
(f) Rudder angle indicator (circuit N).
(g) Propeller revolution indicator (leading shaft-circuit K).
(h) Backing indicator lights for each shaft.
(i) Course-steering transmitter.
(j) Two operating knobs associated with (i).

3.8.2.2 Engine order indicator-transmitter (Starboard-circuit 1MB).- This indicator-transmitter shall provide a means for transmitting starboard engine orders and for receiving acknowledgments of these orders between the ship control indicator-transmitter and the starboard main engine control station. This indicator-transmitter shall contain the mechanism necessary to transmit engine orders (and indicate the orders transmitted), and to indicate the replies (acknowledgments) received. The transmitter shall be positioned by means of an operating knob located beside the instrument window as shown on Drawing S6505-1,430,783. The orders and acknowledgments of a starboard engine order shall be indicated on the right hand instrument. The provisions of 3.5.4.2.1, 3.5.4.2.2, 3.5.4.2.3, 3.5.4.2.4, 3.5.2.2.5, 3.5.2.2.6 3.5.2.2.7, and 3.5.2.2.8 shall apply.

3.8.2.2.1 Terminal markings.- Terminals marked 1MB/T, 1MBB/T, 1MB1, 1MB2, 1MB3, 1MB/I, 1MBB/I, 1MB4, 1MB5, 1MB6, 1MB7, and 1MB8 shall be provided on the terminal strips for the starboard engine order circuit. Terminals 1MB/T and 1MB7 shall be used for connection of the signal push switch. Terminals 1MBB/I and 1MB8 will be used for connection of an audible signal.

3.8.2.3 Engine order indicator-transmitter (Port - circuit 2MB).- This indicator-transmitter shall provide a means for transmitting port engine orders and for receiving acknowledgments of these orders between the ship control indicator-transmitter and the port main engine control station. This indicator-transmitter shall contain the mechanism necessary to transmit engine order (and indicate the

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orders transmitted), and to indicate replies (acknowledgments) received. The transmitter shall be positioned by means of an operating knob located beside the instrument as shown on Drawing S6505-1,430,783. The orders and acknowledgments of a port engine order shall be indicated on the right hand instrument. The provisions of 3.5.2.2.5, 3.5.2.2.6, 3.5.2.2.7, 3.5.2.2.8, 3.5.4.2.1, 3.5.4.2.2, 3.5.4.2.3, and 3.5.4.2.4 shall apply.

3.8.2.3.1 Terminal markings.- Terminals marked 2MB/T, 2MBB/T, 2MB1, 2MB2, 2MB3, 2MB/I, 2MBB/I, 2MB4, 2MB5, 2MB6, 2MB7 and 2MB8 shall be provided on the terminal strips for the port engine order circuit. Terminals 2MB/T and 2MB7 shall be used for connection of the signal push switch. Terminals 1MBB/I and 1MB8 will be used for connection of an audible signal.

3.8.2.4 Rudder angle indicator (circuit N).- The provisions of 3.7.1.4 shall apply.

3.8.2.5 Propeller revolution indicator - (leading shaft - circuit K).- This indicator shall provide a means of indicating the propeller revolutions per minute for the leading shaft. The dial shall be 4 inches in diameter in accordance with Specification MIL-I-983. The provisions of 3.7.1.2.3 and 3.7.1.5.1 shall apply.

3.8.2.5.1 Positioning of pointer.- The pointer shall be positioned by means of a type 23TR6A synchro unit. The synchro shall be adjustable in position and shall be so adjusted in position that the pointer indicates the "O" position when the synchro is at electrical zero.

3.8.2.5.2 Terminal markings.- Terminals marked K, KK, K1, K2, and K3 shall be provided on the terminal strip for the propeller revolution indicator.

3.8.2.6 Backing lights.- The provisions of 3.7.1.5.4 shall apply.

3.8.2.6.1 Terminal markings.- Terminals marked 1K and 1K7 shall be provided for the (outboard) starboard shaft backing light. Terminals marked 2K and 2K7 shall be provided for the (inboard) starboard shaft backing light. Terminals marked 3K and 3K7 shall be provided for the (inboard) port shaft backing light. Terminals marked 4K and 4K7 shall be provided for the (outboard) port shaft backing light.

3.8.2.7 Course - steering transmitter.- This instrument shall provide a means of transmitting (and indicating the orders transmitted) course and steering orders from the ship control indicator-transmitter to the pilothouse steering control console. The provisions of 3.6.2.2.1 shall apply.

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3.8.2.7.1 Steering orders. - A single pointer shall be provided for indicating the steering order. The pointer shall be positioned in conjunction with a type 5G synchro unit by an operating knob below and to the left of the dial window. The type 5G synchro unit shall be adjustable in position and shall be so adjusted in position that the pointer indicates the "CORS" position when the synchro is at electrical zero. Clockwise rotation of the operating knob shall result in clockwise rotation of the pointer.

3.8.2.7.1.1 Operating mechanism. - The operating shaft of the steering order-transmitter shall be fitted with a detent wheel to ensure stability of positioning, and to operate electrical contacts in an audible signal circuit. The detent wheel and the signal contacts shall ensure proper operation (signal contacts shall close and remain closed for a sufficient period of time) of the remote audible signal when the steering order-transmitter operating knob is rotated rapidly from one position to another. The detent wheel shall be provided with a stop at a point corresponding to the center of the space between the two "HARD OVER" orders shown on the steering order section of the instrument dial.

3.8.2.7.1.2 Push switch. - A push switch in accordance with Drawing 9000-S6202-73754 shall be provided on the instrument panel to activate an attention signal at the indicator in the pilothouse and attract attention to a course-steering order. This push switch shall be connected in parallel with detent switch outlined in 3.8.2.7.1.1.

3.8.2.7.2 Course orders. - Indication of the course order shall be provided by means of rotation of the course dial ring, relative to the fixed indication mark shown above "CORS" on figure 10 of Drawing 9000-S6504-73687. The course dial ring shall be positioned in conjunction with a type 5G synchro unit by an operating knob located below and to the right of the dial window. The gearing between the dial ring and the type 5G synchro generator shall be in a 1 to 1 ratio. The gearing between the type 5G synchro generator and the operating knob shall be in a 2 to 1 ratio to allow more accurate setting of the course dial. Clockwise rotation of the operating knob shall result in counterclockwise rotation of the rotating dial.

3.8.2.7.3 Dials. - All dials shall be 4 inches in diameter in accordance with Specification MIL-I-983.

3.8.2.7.4 Illumination control. - A bank of three rheostats with one operating knob shall be provided on the indicator-transmitter as shown on Drawing S6505-1,430,783 for control of instrument illumination. One rheostat shall control the illumination for the course-steering transmitter. A second rheostat shall control the illumination for the port and starboard engine order indicator-transmitter. A third rheostat shall control the illumination for the rudder angle and propeller

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revolution indicators. The power for the illumination circuit shall be taken from the secondary of a transformer whose primary is connected across terminals N and NN. The illumination control shall be provided with an "OFF" position.

3.8.2.7.5 Cable entrance. - Provisions shall be made for the entrance of three type DSGA-3 cables or three type DHFA-3 cables for audible signal circuits for the port and starboard engine order circuits and the course-steering order circuit. Provisions shall also be made for the entrance of type MSCA-37 or type MHFA-37 cables for ships wiring. Entrance holes shall be provided for the smaller diameter cable, with sufficient boss area for possible entrance of the larger diameter cable. The provisions of 3.3.4 shall apply.

3.9 Maintenance parts kits. - The provisions of Specification MIL-I-983 shall apply. A kit of maintenance parts shall consist of the following:

(a) An instrument of each type included in the equipment, except for those instruments which are Government furnished. (A Government furnished magnesyn compass repeater with a contractor furnished means of illumination shall be included in the steering control console maintenance parts kits.)
(b) Parts frequently requiring renewal under operating conditions, or subject to considerable wear in operation.
(c) Special tools and accessories necessary for care, maintenance and adjustments to be made aboard ship. The use of special tools shall be held to a minimum.
(d) Adapter cables, in quantity as necessary, for connecting an instrument to the console when the instrument is removed from the console for servicing. Cables shall be 10 feet in length.
(e) One cable assembly in general accordance with Drawing S2407-533062. (Applies only to steering control consoles which contain a magnesyn compass indicator.)

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection procedures. - For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

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4.2 General. - The sampling procedures for the various tests to be performed on the equipment shall be as follows: (The attention of the manufacturer is invited to the applicable paragraphs of Specification MIL-I-983 for information concerning these tests such as cost, where conducted, failures, test conditions, etc.)

4.2.1 Qualification tests at a Government laboratory. - Qualification tests shall be conducted at a Government laboratory designated by the Bureau of Ships. One each of the following ship control combination instruments (or equivalent except for another permissible enclosure material) shall be subjected to qualification tests.

Ship Control Indicator, Type IC/12B02
Ship Control Indicator-Transmitter, Type IC/14S03

Tests shall be performed, in general, in the order listed:

Test	Requirement paragraph	Test paragraph
General inspection	3.2.1	4.3.1
Supply line voltage and frequency	3.2.1	MIL-I-983
Radio frequency noise interference	3.2.1	MIL-I-983
Audible noise (Airborne only)	3.2.1	MIL-I-983
Oscillations	3.4.10	4.3.3
Temperature and humidity	3.2.1	MIL-I-983
Inclination	3.2.1	MIL-I-983
Dielectric strength	3.2.1	MIL-I-983
Insulation resistance	3.2.1	MIL-I-983
Instrument illumination	3.4.8	4.3.4
Accuracy	3.4.9	4.3.2
Enclosure	3.2.1	MIL-I-983
Salt spray	3.2.1	MIL-I-983
Vibration	3.2.1	MIL-I-983
Shock	3.2.1	MIL-I-983

4.2.1.1 Manufacturers qualifying for types IC/12B02 and IC/14S03 shall be considered as qualified for all types of equipment covered by this specification.

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4.2.2 Preliminary.- The first production unit of each type equipment covered in the contract or order shall be subjected to the following preliminary tests at a laboratory designated by the bureau or agency concerned. These tests will be required in the contract or order only where the successful bidders products have not passed the qualification tests listed in 4.2.1. These tests shall be performed in general, in the order listed:

Test	Requirement paragraph	Test paragraph
General inspection	3.2.1	4.3.1
Supply line voltage and frequency	3.2.1	MIL-I-983
Radio frequency noise interference	3.2.1	MIL-I-983
Audible noise (Airborne only)	3.2.1	MIL-I-983
Oscillations	3.4.10	4.3.3
Temperature and humidity	3.2.1	MIL-I-983
Inclination	3.2.1	MIL-I-983
Dielectric strength	3.2.1	MIL-I-983
Insulation resistance	3.2.1	MIL-I-983
Instrument illumination	3.4.8	4.3.4
Accuracy	3.4.9	4.3.2
Enclosure	3.2.1	MIL-I-983
Salt spray	3.2.1	MIL-I-983
Vibration	3.2.1	MIL-I-983
Shock	3.2.1	MIL-I-983

4.2.3 Inspection tests at the place of manufacturer.- A sample number of equipments shall be selected in accordance with 4.2.3.1.2, 4.2.3.1.3, and 4.2.3.1.4 and each of the selected equipments shall be subjected to the following tests at the place of manufacture, except that group C tests shall be conducted at a laboratory designated by the bureau or agency concerned. The tests shall be performed in general, in the order listed and shall be conducted or witnessed by the Government inspector.

Test	Requirement paragraph	Test paragraph
Group A tests		
General inspection	3.2.1	4.3.1
Dielectric strength	3.2.1	MIL-I-983
Insulation resistance	3.2.1	MIL-I-983

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Test	Requirement paragraph	Test paragraph
Group B tests		
Supply line voltage and frequency variation	3.2.1	MIL-I-983
Oscillations	3.4.10	4.3.3
Accuracy	3.4.9	4.3.2
Group C tests		
Radio frequency noise interference	3.2.1	MIL-I-983
Audible noise (Airborne only)	3.2.1	MIL-I-983
Temperature and humidity	3.2.1	MIL-I-983
Inclination	3.2.1	MIL-I-983
Enclosure	3.2.1	MIL-I-983
Salt spray	3.2.1	MIL-I-983
Vibration	3.2.1	MIL-I-983
Shock	3.2.1	MIL-I-983

4.2.3.1 Sampling for inspection tests.-

4.2.3.1.1 Inspection lot.- All equipment of the same type presented for a Government inspection at one time shall be considered a lot. The lot may include the entire contract quantity or it may be the production of any convenient time period at the choice of the Government inspector. Prior to presenting the lot for inspection, the contractor shall have subjected each equipment to the group A and B tests and shall include in the lot presented for Government inspection only those equipments that have passed these tests.

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4.2.3.1.2 Sampling for group A tests. - A sample of equipment shall be selected from each inspection lot by the Government inspector in accordance with table VIII and shall be subjected to each of the group A tests specified in 4.2.3 with lot acceptance based on the sampling procedures of Standard MIL-STD-105.

Table VIII - Sampling for group A tests
AQL (approx.) = 1.5 percent defective.

Number of equipments in inspection lot	Number of equipments in sample	(Number of equipments nonconforming on any group A test)	
		Acceptance number	Rejection number
7 and under	All	---	---
8 to 15	7	0	1
16 to 40	10	0	1
41 to 110	15	0	1
111 to 300	25	1	2
301 to 500	35	1	2
501 and over	50	2	3

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4.2.3.1.3 Sampling for group B tests. - A sample of equipments shall be selected from each inspection lot by the Government inspector in accordance with table IX and shall be subjected to each of the group B tests specified in 4.2.3 with lot acceptance based on sampling procedures of Standard MIL-STD-105.

Table IX - Sampling for group B tests.

Number of equipments in inspection lot	Number of equipments in sample	(Number of equipments nonconforming on any group B test)	
		Acceptance number	Rejection number
3 and under	All	---	---
4 to 15	3	0	1
16 to 40	5	0	1
41 to 110	7	0	1
111 to 300	10	0	1
301 to 500	15	1	2
501 and over	25	2	3

4.2.3.1.4 Sampling for group C tests. - Group C tests will be required by the bureau or agency concerned when the basic design of the equipment or the material of a vital part has been changed. One complete equipment shall be selected by the Government inspector and subjected to each of the group C tests specified in 4.2.3 with acceptance based on 4.2.3.1.6.

4.2.3.1.5 Acceptance - group A and B tests. - The results of each test shall be compared with this specification. Failure to conform to this specification for any test shall be counted as a defect and the equipment shall be rejected. If the number of such nonconforming equipments in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Rejected lots may be offered again for Government inspection provided all equipments in the lot have been retested by the contractor for the test(s) causing rejection and all noncompliance corrected.

4.2.3.1.6 Acceptance - group C tests. - The results of each test shall be compared with this specification. In the event of failure to conform to this specification for any group C test, the contractor shall correct the cause of failure on future production units and repair the deficiency in all equipments produced on the contract or order.

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4.3 Test procedures. -

4.3.1 General inspection. - The general inspection shall be conducted in accordance with Specification MIL-I-983. The operating test shall consist of the connection of, and testing as a system, instruments assembled and connected in accordance with the loading characteristics of synchro units as specified in 3.4.6.1. The system shall be set up in accordance with the maximum loading limitations specified in 3.4.6.1. Instruments shall be used as required to make up such systems, and allowances shall be made for nonagreement of instrument dials among various types of instruments which may be assembled to comprise a test system. Instruments conforming a complete system for the operating test shall be connected with all associated auxiliary equipment, such as bells and buzzers, and operated at least 20 times without failure of a single item of equipment. Failure of any item of equipment shall require rectification of the item of the equipment by the contractor, and a repetition of the operating test.

4.3.2 Accuracy. - All indicator pointers shall follow the transmitter synchro generators within the requirements of 3.4.9. Readings shall be taken and a curve plotted for each 15 degrees, clockwise and counterclockwise, around the complete dial.

4.3.3 Oscillation. - When the electrical supply is disconnected and the transmitter synchro generator rotor is off-set mechanically and locked at positions of 5, 10, 20, 90, and 180 degrees out of synchronism with the indicators, the elapsed time between the connection of the electrical supply and the time the indicator pointer comes to rest at the position synchronous with the transmitter, shall comply with 3.4.10.

4.3.4 Dial illumination. - Final approval of the dial illumination shall be based on critical examination in darkness, and when illuminated by external light of an intensity representing that prevalent in normal shipboard installations. This evaluation shall give full consideration to the useful angle of view, to the daylight contrast and general legibility of the pointers and dial markings, and to the uniformity, contrast, legibility, and freedom from distracting glare of the instrument dial when illuminated through use of its internal illumination, and viewed at normal and abnormal angles of view.

4.3.5 Other examination and tests. - Such other examination and tests as the inspector may consider necessary shall be made in order to determine the suitability for Naval use of the equipment under test.

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5. PREPARATION FOR DELIVERY

5.1 Basic equipment and accessories. - The cleaning, preservation, packaging, and packing shall be as specified below. The equipment shall be prepared for immediate use, storage, stock or overseas shipment as specified (see 6.1).

5.1.1 Immediate use. - Equipment and accessories destined for shipment to domestic shipyards for immediate installation or use shall be cleaned, preserved, packaged, and packed as follows:

5.1.1.1 Cleaning, preservation, and packaging. - The cleaning, preservation and packaging shall be in accordance with the manufacturers commercial practice.

5.1.1.2 Packing. - Equipment and accessories shall be packed in accordance with the latest edition of the Consolidated Freight Classification Rules in effect on the date of invitation to bid. Shipping containers shall be of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carrier, at the lowest rate, to the point of delivery.

5.1.2 Storage, stock, or overseas shipment. - Equipment and accessories destined for storage, stock, or overseas shipment shall be cleaned, preserved, packaged, and packed as follows:

5.1.2.1 Cleaning, preservation, and packaging. - Equipment and accessories scheduled for storage, stock or overseas shipment shall be cleaned by method C-1 and packaged by method II in accordance with Specification MIL-P-116. No contact preservative is required.

5.1.2.2 Packing. -

5.1.2.2.1 For storage or stock. - Equipment and accessories (except for method IIa packages) destined for ultimate installation on a specific ship but temporarily stored or for stock shall be packed in wood cleated fiberboard, cleated plywood, nailed wood, corrugated or solid fiberboard boxes or nailed wood crates conforming to Specification NN-B-591, NN-B-601, NN-B-621, LLL-B-631, or LLL-B-636 or MIL-C-6057, respectively. Closure of the fiberboard boxes shall be made with adhesive, metal fastenings, tape or a combination of these methods, in such a manner that the closures of the boxes shall not break or open when the box is tested in accordance with the applicable container specification. The gross weight of wood boxes shall not exceed 150 pounds unless the weight of the individual item exceeds this amount. The gross weight of fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

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5.1.2.2.2 For overseas shipment. - Equipment and accessories except for method IIa packages destined for overseas shipment shall be packed in wood cleated plywood, nailed wood, fiberboard boxes or nailed wood crates conforming to style A or B of Specification JAN-P-105, Specification JAN-P-106, symbol V3c or V3s of Specification JAN-P-108 or Specification JAN-P-104, respectively. Plywood shall conform to type A or B, condition I of Specification JAN-P-139. Boxes shall be lined with a sealed waterproof case liner conforming to Specification MIL-L-10547. Seams and closures shall conform to appendix of liner specification. Shipping containers shall be closed and strapped in accordance with the appendix of the applicable container specification. Flat steel strapping shall conform to class A or B of Specification QQ-S-781. The gross weight of wood boxes shall not exceed 150 pounds unless the weight of the unit item exceeds this amount. The gross weight of fiberboard boxes shall not exceed 70 pounds.

5.2 Repair parts.

5.2.1 Cleaning, preservation, and packaging. - Repair parts shall be cleaned, preserved, and packaged in accordance with Specification MIL-P-116, using table I of Specification JAN-P-658 as a guide.

5.2.2 Packing. - Where onboard repair parts and tools are destined for a ship that is not fitted with bin or drawer type stowage (see 6.1), the onboard repair parts and tools shall be packed in type M metal boxes conforming to Specification MIL-B-233 and shall be overpacked in shipping containers conforming to Specification NN-B-621 or in a nailed wood crate conforming to Specification JAN-P-132. Where onboard repair parts and tools are destined for a ship that is fitted with bin or drawer type stowage (see 6.1), both onboard and stock repair parts shall be packed in wood cleated plywood, nailed wood, corrugated or solid fiberboard boxes conforming to Specification NN-B-601, NN-B-621, LLL-B-631 or LLL-B-636, respectively. Fiberboard boxes shall conform to the special requirements of the applicable box specification. Closure of the fiberboard boxes shall be made with adhesive, metal fastenings, tape or a combination of these methods, in such a manner that the closures of the boxes shall not break or open when the box is tested in accordance with the applicable container specification. The gross weight of wood boxes shall not exceed 150 pounds unless the weight of the individual item exceeds this amount. The gross weight of fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.3 Publications. -

5.3.1 Packaging. - Instruction books and repair parts list shall be packaged in accordance with Specification MIL-B-15071.

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5.3.2 Packing. - When specified in the contract or order, two copies of the instruction book shall be packed within the shipping container holding the main unit. Bulk quantities of instruction books shall be packed in accordance with Specification MIL-B-15071. One copy of the repair parts list shall be packed in each type M box.

5.4 Marking. -

5.4.1 Interior containers. - In addition to the requirements specified in Standard MIL-STD-129, each interior package shall be clearly and legibly marked with the electrical and operational characteristics (as applicable).

5.4.2 Exterior containers. - In addition to the marking specified in 5.4.1 and any special marking required by the contract or order, shipments shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. - Procurement documents shall specify the following information in the contract or order:

(a) Title, number, and date of this specification.
(b) Type of equipment required (see 1.2).
(c) Quantity of items required.
(d) Supply Demand Control Center for repair parts (see 3.2).
(e) Quantity of maintenance parts (see 3.9).
(f) Quantity of instruction books (see 3.2).
(g) Whether or not preliminary tests are required (see 4.2.2).
(h) Whether equipment is to be prepared for immediate use, storage, stock or overseas shipment (see 5.1).
(i) Type of stowage for onboard repair parts (see 5.2.2).
(j) Whether or not instruction books are to be packed with the equipment (see 5.3.2).

6.2 In the procurement of products requiring qualification, the right is reserved to reject bids on products that have not been subjected to the required tests and found satisfactory for inclusion on the Military Qualified Products List. The attention of suppliers is called to this requirement, and manufacturers are urged to communicate with the Bureau of Ships, Navy Department, Washington 25, D.C., and arrange to have the products that they propose to offer to the Army, the Navy, or the Air Force, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Chief of the Bureau of Ships, Navy Department, Washington 25, D.C.

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6.3 Nomenclature. - The name of parts shall conform to "Item Description Guides for Supply Cataloging" published by the Munitions Board Cataloging Agency.

6.4 The packaging, packing and marking specified herein apply only to direct shipment to the Government and are not intended to apply to contract or orders between the manufacturer and the prime contractor.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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PREVIOUS EDITION IS OBSOLETE.

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